



POLITECNICO DI BARI

Department of Electrical and Information Engineering

**MASTER'S DEGREE PROGRAMME IN
TELECOMMUNICATIONS ENGINEERING**

RADIO PROPAGATION PROJECT

TITLE

**RADIO COVERAGE SIMULATION FOR THE FESTA DEI SANTI
MEDICIIN BITONTO WITH DYNAMIC DRONE-BASED
SUPPORT**

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Academic Year 2025-2026

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INTRODUCTION

Radio communications with mobile users represent one of the most relevant areas in modern telecommunications. However, in real propagation scenarios, link quality can be strongly affected by obstacles, the structure of the urban environment, terrain irregularities, and the operating conditions of the system. These factors may introduce attenuation and degradation of the received signal, meaning that radio service continuity is not always guaranteed.

This project originates from a real experience gained while providing health-care assistance as a volunteer with the Italian Red Cross (CRI) during the Festa dei Santi Medici in Bitonto. This event is characterized by a large number of attendees and by a procession that crosses different areas of the city, making reliable communication between the Advanced Medical Post (PMA) and the operational teams distributed along the route essential. During the service, it was possible to observe that the radio signal from the central station was not always uniform throughout the city, with possible critical issues in areas more heavily shielded by buildings or farther from the fixed station.

Starting from this observation, the project proposes a radio coverage simulation along the procession route, considering the PMA as the fixed transmitting station and the procession as a mobile user to be tracked over time. The goal is to evaluate the quality of the direct link between the central station and the mobile team and, in the sections where the received signal drops below a predefined threshold, to introduce a drone as a radio support node.

The simulation was developed in MATLAB, using cartographic data related to the urban area of Bitonto and adopting the Longley-Rice model as the propagation model. This model makes it possible to estimate radio propagation in scenarios that are more realistic than simple free-space propagation, taking into account the effects related to distance, terrain, and non-ideal link conditions.

The first chapter introduces the reference application context, the main theoretical concepts related to the radio link, the link budget, and the Longley-Rice propagation model used in the simulations. The second chapter describes the MATLAB modeling of the scenario, with particular attention to the definition of the PMA, the procession route, the antennas, and the simulation parameters. The third and final chapter analyzes the drone-based support strategy, describing the activation criteria, the search for the optimal position, the orientation of the directional antenna, possible future developments of the system, and the analysis of the results.

Chapter 1

Theoretical background and application scenario

1.1 Operational context

The Festa dei Santi Medici in Bitonto is an important religious and civic event, during which the presence of a large number of participants requires careful organization of health-care assistance and operational coordination activities. In this context, radio communication plays an essential role, as it allows the PMA to maintain contact with the teams present along the procession route.

The considered scenario is therefore characterized by a fixed transmitter, located at the operations center, and a mobile receiver, representing the health-care team following the procession. The main problem is to assess whether the direct radio link can provide a sufficient received power level along the entire route.

Radio propagation in an urban environment is affected by the presence of buildings, narrow streets, obstacles, reflections, diffractions, and possible non-line-of-sight conditions between transmitter and receiver. These phenomena may generate shadow zones, where the received signal is lower than the level required to ensure stable communication.

In the case under study, the procession crosses a portion of the city where the link with the PMA may vary according to distance and to the geometric configuration of the environment. For this reason, it is useful to build a simulation model capable of estimating the received power along the route and identifying the most critical points. In this context, the use of the link budget is extremely useful.

1.2 Link budget

The link budget represents the power balance of a radio link. It makes it possible to estimate the received power from the transmitted power, antenna gains, and the losses experienced by the signal during propagation. In general form, the link budget can be expressed as:

$$P_r = P_t + G_t + G_r - L_p - L_{misc} \quad (1.1)$$

where:

- P_r is the received power;

- P_t is the transmitted power;
- G_t is the gain of the transmitting antenna;
- G_r is the gain of the receiving antenna;
- L_p represents free-space propagation losses (Free Space Loss);
- L_{misc} represents any additional system losses.

Using quantities expressed in dB, dBm, and dBi, the relationship becomes:

$$P_r[\text{dBm}] = P_t[\text{dBm}] + G_t[\text{dBi}] + G_r[\text{dBi}] - L_{FSL}[\text{dB}] - L_{misc}[\text{dB}] \quad (1.2)$$

In this project, the calculation of the received power is performed using the radio simulation functions available in MATLAB [1]. In particular, the *sigstrength* command is used [2]:

```
sigstrength(rx, tx, pm);
```

However, the concept of link budget remains fundamental for interpreting the obtained results and understanding which factors determine the received signal level.

1.3 Free-space propagation

A first theoretical reference model is free-space propagation, which considers an ideal link without obstacles between transmitter and receiver. This model, used in the link budget calculation, is represented by the following relationship:

$$L_{FSPL} = 32.44 + 20 \log_{10}(f_{MHz}) + 20 \log_{10}(d_{km}) \quad (1.3)$$

where f_{MHz} is the frequency expressed in MHz and d_{km} is the distance between transmitter and receiver expressed in kilometers [3]. This model is useful as a theoretical reference, but it is not sufficient to fully describe a real urban scenario.

1.4 Longley-Rice model

To overcome the limitations of free-space propagation alone, the Longley-Rice model, also known as the *Irregular Terrain Model* (ITM), was adopted in the simulation. This model makes it possible to estimate propagation loss along a terrestrial radio link while considering more realistic conditions than the ideal case described by Free Space Loss.

In free-space propagation, it is assumed that there are no obstacles between transmitter and receiver and that the signal is attenuated only as an effect of distance. This assumption is useful as a theoretical reference, but it is not sufficient when the link takes place in a real environment, where propagation may be influenced by terrain morphology and by the presence of elements that modify the path of the electromagnetic wave.

The Longley-Rice model combines deterministic and statistical contributions to describe signal attenuation. In particular, it considers distance-related loss, diffraction effects, possible line-of-sight or non-line-of-sight radio conditions, and propagation channel variability. For this reason, it is suitable for the study of radio links in scenarios where the received power does not depend only on the geometric distance between transmitter and receiver [4].

In this project, the Longley-Rice model is used as the propagation model within the MATLAB environment. It is defined using the command:

```
pm = propagationModel("longley-rice")
```

Once defined, the model is used to estimate the received power level at the different points of the procession route. In this way, the evaluation of radio coverage is not based only on the distance from the PMA, but also takes into account a more realistic description of the propagation channel [5].

The use of Longley-Rice therefore provides, for each position of the procession, a received power value that is more representative of the considered operational scenario. These values form the basis for identifying the sections where the direct link with the PMA is less favorable and for evaluating the effectiveness of the subsequent drone-based support strategy.

1.5 Reception threshold

To determine when the direct radio link can be considered critical, a minimum received power threshold was introduced. In the implemented model, when the signal received by the procession falls below this threshold, the system activates the support drone.

Indicating with $P_{r,dir}$ the received power in the direct link between the PMA and the procession, the activation condition can be expressed as:

$$P_{r,dir} < P_{threshold} \quad (1.4)$$

where $P_{threshold}$ represents the minimum level required to consider the link sufficiently reliable. The chosen value was -65 dBm.

The use of a drone makes it possible to introduce a mobile radio node capable of changing its position according to link conditions. The drone can therefore intervene in the sections where the direct signal between the PMA and the procession is insufficient, searching for a position from which it is possible to maintain a good link with both the central station and the mobile team.

In the project, the drone is considered as a support node equipped with a directional antenna. This choice makes it possible to concentrate the signal in the direction of the procession, potentially improving link quality in the critical points.

Chapter 2

Simulation modeling and implementation

2.1 Simulation environment

The simulation was developed in the MATLAB environment using tools dedicated to the modeling of radio scenarios. In particular, functions were used to define transmitting and receiving sites, visualize the urban environment, and calculate the received power through a propagation model [6].

The urban environment was reconstructed using cartographic data related to the city of Bitonto, imported from OpenStreetMap. In this way, it was possible to represent the presence of buildings and place the simulation within the considered real scenario.

2.2 Walkie-talkie antenna definition

For the modeling of the radio link through walkie-talkies, a communication between the PMA, assumed as the transmitting node, and the team on foot following the procession, assumed as the receiving node, was considered through the MATLAB `txsite` function [7].

As a reference for defining the radio parameters, a professional portable device was considered, representative of the devices that can be used in Civil Protection and territorial health-care assistance contexts. In particular, the *Motorola MOTOTRBO R5* model was considered as an example. According to the manufacturer's technical datasheets, this device has a maximum transmit power of 5 W in the VHF configuration [8]. For this reason, the following transmit power was assumed in the simulations:

$$P_t = 5 \text{ W}$$

The considered operating frequency is:

$$f_c = 160 \text{ MHz}$$

The antenna used for both terminals is a dipole designed at the indicated center frequency. In MATLAB, the antenna was defined using the following code:

```
fc = 160e6;  
antenna = design(dipole,fc);
```



```
antenna.Length = 0.15;  
antenna.Width = 0.01;
```

The dipole was then modified by setting a length of 0.15 m and a width of 0.01 m. These values allow the physical antenna associated with the walkie-talkie devices used in the operational scenario to be represented in a simplified manner.

In the considered model:

- the transmitting node, indicated as TX, corresponds to the PMA;
- the receiving node, indicated as RX, corresponds to the **team on foot following the procession**;
- both nodes use the same dipole antenna;
- the operating frequency is 160 MHz.

Figure 2.1 shows the geometry of the dipole antenna used in the model, while Figure 2.2 shows the corresponding three-dimensional radiation pattern.

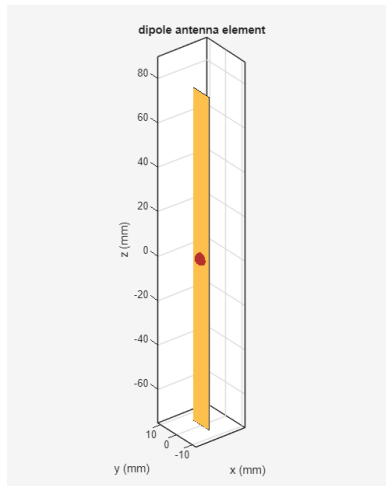


Figure 2.1: Geometry of the dipole antenna used for the walkie-talkie link.

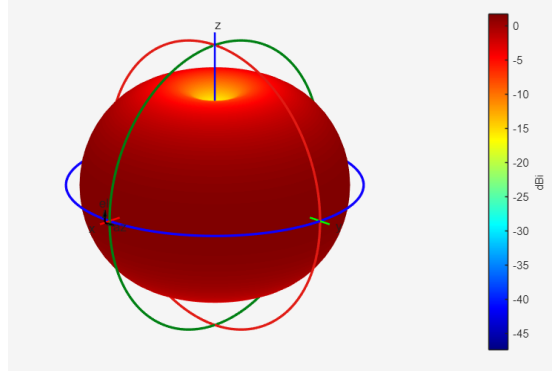


Figure 2.2: Three-dimensional radiation pattern of the dipole antenna at the frequency of 160 MHz.

This configuration makes it possible to represent the radio link between the PMA and the team on foot during the progression of the procession, considering propagation that depends on the radiating characteristics of the antenna and on the relative position between transmitter and receiver.

2.3 Procession route

The procession was represented by an ordered sequence of geographic coordinates expressed in latitude and longitude. Each row identifies a significant point of the route.

The structure of the matrix used in the MATLAB code is the following:

```
coords = [
    lat1 , lon1 ;
    lat2 , lon2 ;
    ...
    latN , lonN
];
```

where the first column contains the latitudes and the second column contains the longitudes of the points belonging to the procession route.

Starting from the coordinate matrix, the vectors containing the latitude and longitude of the route points are extracted. Then the distance between consecutive points is calculated, obtaining a measure of the separation between the various sections of the trajectory.

The discretization step is chosen as the minimum distance between two consecutive points. In this way, the entire route is sampled with a sufficiently dense resolution even in the shortest sections.

To obtain a physically interpretable estimate of the adopted step, the minimum

distance is converted into meters by considering the mean Earth radius equal to:

$$R = 6371000 \text{ m}$$

Subsequently, for each pair of consecutive coordinates, the code generates a set of intermediate points through linear interpolation. This operation makes it possible to discretize the entire procession route uniformly.

The interpolated points are stored in the final matrix `FinalRoute`, which represents the complete trajectory used in the radio model.

In the considered system, each point of the `FinalRoute` matrix represents a successive position of the mobile receiver, namely the team on foot following the procession. In this way, the received power is evaluated along the entire route and not only at the main points of the original track, allowing any radio signal attenuation zones to be identified with greater precision.

The obtained discretized route is shown in Figure 2.3. Each marker represents a successive position assumed by the mobile receiver along the procession route.

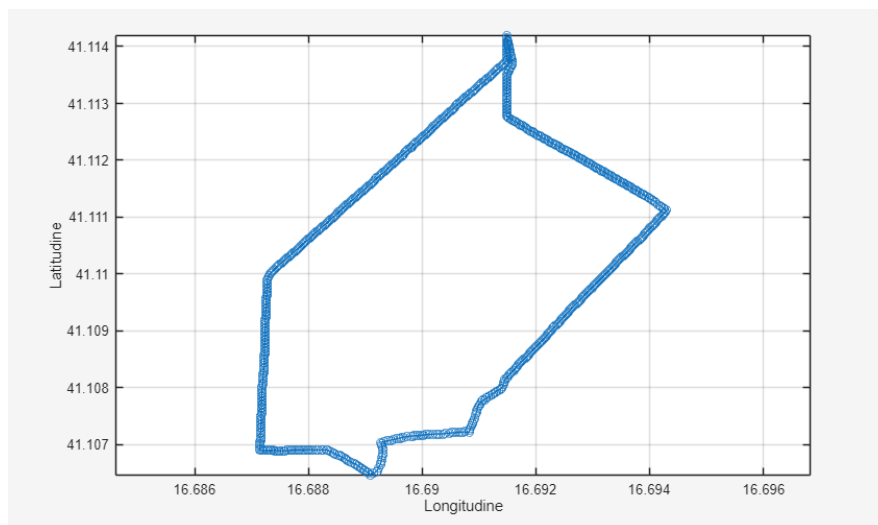


Figure 2.3: Discretized procession route obtained from the geographic coordinates.

To highlight the result of the discretization, Figure 2.4 shows a zoom on a section of the route. It can be observed that the interpolated points are regularly distributed along the considered segment, allowing a more continuous evaluation of the mobile receiver position.

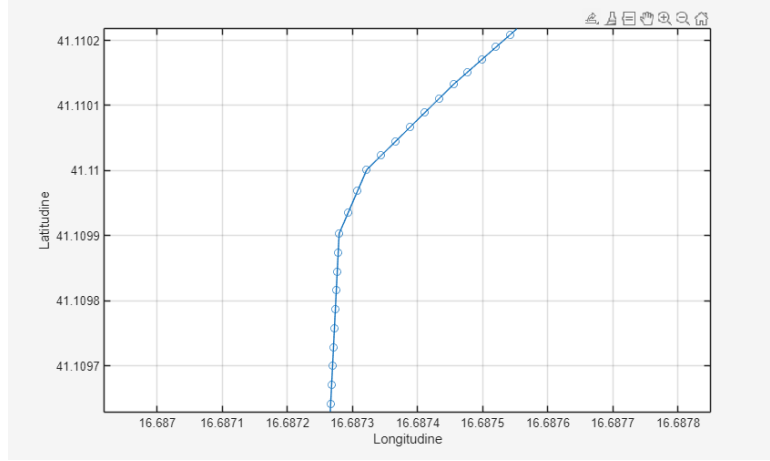


Figure 2.4: Detail of the discretized route.

2.4 Mobile receiver

The health-care team following the procession was modeled as a mobile receiver, whose position varies along the discretized route described in the previous section.

The receiver is defined through an `rxsite` object [9], to which the geographic coordinates contained in the `FinalRoute` matrix are assigned at each iteration. In this way, for each point of the route, the model updates the receiver position while keeping the other characteristics of the system constant.

The height of the receiving antenna was set to:

$$h_{RX} = 1.5 \text{ m}$$

This value represents a height compatible with the use of a portable radio device by an operator on foot. In addition, the same dipole antenna defined for the walkie-talkie system is associated with the receiver.

For each position of the `FinalRoute` matrix, the code calculates the received power through the `sigstrength` function, using the transmitter `tx` and the previously defined propagation model `pm`. The obtained result is saved in the `ss` variable, which represents the received signal level at the considered point.

The algorithm therefore performs a complete scan of the route: at each step, a new receiver is generated at the current position, the received power is calculated, and the value is displayed together with the corresponding coordinates.

In this way, a profile of the received power along the entire procession route is obtained. This profile makes it possible to identify the sections where the radio link between the PMA and the team on foot is adequate and those where the signal is more strongly attenuated, possibly requiring the support of auxiliary systems such as drones

or additional radio relay points.

2.5 Coverage map

In addition to the point-by-point calculation of the received power along the procession route, a radio coverage map of the considered area was generated using the coverage command [10]. This representation allows the spatial distribution of the signal transmitted by the PMA to be displayed graphically.

The coverage map makes it possible to identify the zones where the received power level is higher and those where the signal is more strongly attenuated. In particular, areas characterized by lower received power values may correspond to zones shielded by buildings, urban obstacles, or less favorable propagation conditions.

The representation was obtained using the transmitter defined in the model, the selected propagation model, and the antenna associated with the PMA. The result is a signal distribution over the geographic area of interest, useful for visually comparing the procession route with the radio coverage zones.

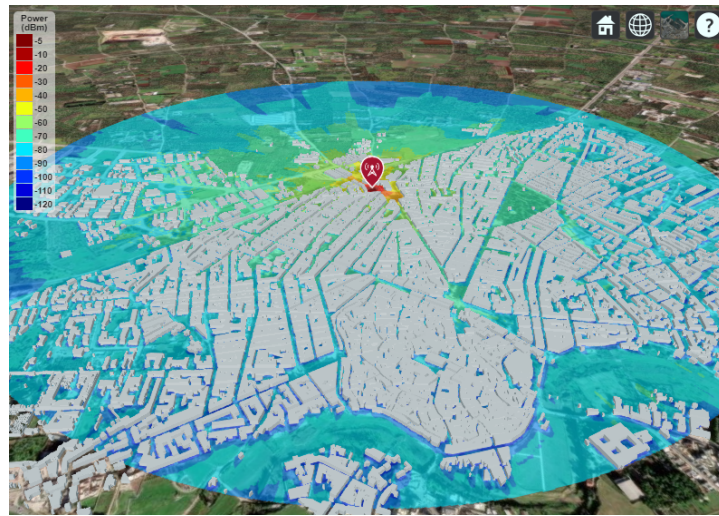


Figure 2.5: Radio coverage map generated from the PMA in the procession area.

Figure 2.5 shows the coverage map obtained from the simulation. The color scale indicates the received power level expressed in dBm: values closer to 0 dBm indicate a stronger signal, while lower values indicate a weaker signal. The map therefore represents a support tool for interpreting the simulation results, since it makes it possible to compare the position of the team on foot along the route with the spatial distribution of the radio signal.

Chapter 3

Drone-based support strategy

In the simulated system, it can be observed that, at several points of the route, the signal received by the team on foot has rather low values. In particular, the worst measured value is equal to -84.5 dBm, measured at the following coordinates:

- latitude = 41.1065
- longitude = 16.6891

This result highlights the presence of critical zones along the procession route, where the direct radio link between the PMA and the team on foot is strongly degraded.

To improve the performance of the communication system, the introduction of a drone acting as a *relay node*, namely an intermediate repeater node between the PMA and the moving team, was considered. The drone was equipped with two antennas: a directional Yagi-Uda antenna, used for transmission with a power of 2 W, and a second receiving dipole antenna, modeled with the same characteristics as the antenna assigned to the team on foot. In particular, the receiving dipole installed on the drone was oriented downward by imposing a 180° rotation with respect to the z -axis, so as to direct the radiation pattern toward the area below affected by the procession.

In order to optimize the drone energy consumption, the support service is not kept continuously active, but is enabled only when the received signal level drops below a predefined threshold, assumed equal to -65 dBm. When this condition occurs, the system searches for a position reachable by the drone that allows the radio link toward the procession to be improved.

3.1 Directional antenna definition

The dimensions of the Yagi-Uda antenna, modeled using the `yagiUda` object [11], were chosen as a function of the wavelength associated with the operating frequency. Since the frequency used in the simulation is equal to 160 MHz, the wavelength is:

$$\lambda = \frac{c}{f_c} = \frac{3 \cdot 10^8}{160 \cdot 10^6} \simeq 1.875 \text{ m}$$

The Yagi-Uda elements were therefore dimensioned as fractions of the wavelength. The reflector was chosen equal to 0.98 m , that is approximately 0.52λ , slightly longer than

the driven element. The driven element has a length of 0.90 m , that is approximately 0.48λ , a value close to the typical length of a resonant dipole. The directors were instead chosen equal to 0.83 m , that is approximately 0.44λ , making them slightly shorter than the driven element, as normally occurs in Yagi-Uda antennas.

The spacings between the elements were also chosen as fractions of the wavelength. In particular, the distance between the reflector and the driven element is 0.30 m , approximately 0.16λ , while the spacing between the directors is 0.28 m , approximately 0.15λ . These values make it possible to obtain a compact configuration, suitable for modeling a radio node installed on a drone, while maintaining a structure consistent with the directional operation of the Yagi-Uda antenna [12].

Figure 3.1 shows the radiation pattern of the implemented antenna.

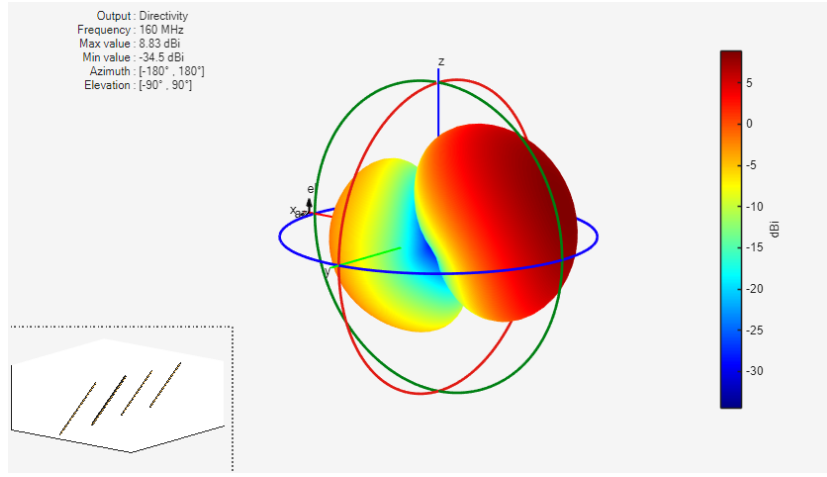


Figure 3.1: Radiation pattern of the drone directional antenna [13].

3.2 Kinematic constraints

To make the simulation more realistic, the drone movement was constrained by its maximum speed. Similarly, the procession was modeled as a mobile user moving at a low speed compatible with the real scenario.

The time required by the procession to move between two successive points can be expressed as:

$$t_{step} = \frac{d_{step}}{v_{procession}} \quad (3.1)$$

where d_{step} is the distance between two consecutive points of the route and $v_{procession}$ is the procession speed. During this interval, the drone can travel at most:

$$d_{drone} = v_{drone} \cdot t_{step} \quad (3.2)$$

This distance defines the maximum radius within which the drone can search for a new position at each simulation step.

3.3 Drone coverage scenarios

The drone was designed to operate in three different scenarios:

- Received signal < -65 dBm, with the drone not yet deployed;
- Received signal < -65 dBm, with the drone already deployed;
- Received signal > -65 dBm, but with the received signal below the threshold value until a few previous steps.

3.3.1 Insufficient coverage with the drone not yet deployed

The first considered scenario concerns the case in which the received signal power of the team on foot is lower than the predefined minimum threshold and the drone has not yet been deployed. In the code, this condition is identified through the check:

$$ss < -65 \text{ dBm}$$

where ss represents the power received by the team terminal. When this value falls below the threshold, the direct link between the PMA and the team is considered no longer adequate, making the intervention of the drone necessary as a support node for communication.

The first part of the logical check therefore verifies whether the drone is not yet active or whether it is in the return phase. In the latter case, deployment and initialization operations are performed again.

```
if ss < -65
    k = 0;
    if drone_attivo == false || drone_rientro == true

        drone_attivo = true;
        drone_rientro = false;
```

In this phase, the counter k , used in the following scenarios to evaluate the drone return, is reset. Subsequently, the drone is set to the active state, while any return condition is disabled. In this way, the system forces the drone deployment whenever

the received power drops below the threshold and the drone is not already in a useful operating condition.

Once activated, the drone is modeled in MATLAB through a `txsite` object, that is, as a transmitting site. The initial position of the drone is defined by the coordinates stored in the `CoordinateDrone` variable, while the antenna height is set to 60 m. The transmission frequency is fixed at 160 MHz, consistently with the rest of the simulation, and the transmitted power is assumed equal to 2 W. The previously defined Yagi-Uda is used as the transmitting antenna:

```
DroneTx = txsite( ...
    Name="DroneTX", ...
    Latitude=CoordinateDrone(1), ...
    Longitude=CoordinateDrone(2), ...
    AntennaHeight=60, ...
    TransmitterFrequency=160e6, ...
    TransmitterPower=2, ...
    Antenna=antennaYagi);
```

After creating the transmitting node, the initial geometric direction toward the receiver, namely toward the current position of the team on foot, is calculated. In particular, the azimuth between the drone and the receiving site, the geodetic distance between the two points, and the height difference between the respective antennas are determined:

```
az0 = azimuth(DroneTx.Latitude, DroneTx.Longitude, ...
    sites(i).Latitude, sites(i).Longitude);

d = distance(DroneTx.Latitude, DroneTx.Longitude, ...
    sites(i).Latitude, sites(i).Longitude, ...
    wgs84Ellipsoid);

dh = sites(i).AntennaHeight - DroneTx.AntennaHeight;

el0 = atan2d(dh, d);
```

The initial elevation angle is obtained using the `atan2d` function, considering the ratio between the height difference dh and the horizontal distance d . In this way, a first estimate of the orientation of the directional antenna toward the receiver is obtained. However, since radio propagation in an urban environment does not depend only on the

direct geometry between transmitter and receiver, but also on the presence of obstacles, reflections, and attenuations introduced by buildings, the geometric pointing is not assumed as definitive.

For this reason, a search is carried out for the pointing angle that maximizes the received power. The procedure is divided into two phases: a first coarse search and a subsequent finer search. In the coarse search, different azimuth and elevation values around the initial direction are explored:

```
az_coarse = az0 -180:10:180;
el_coarse = el0 + (-75:5:75);
```

```
Pr_best = -Inf;
az_best = az0;
el_best = el0;
```

For each azimuth and elevation combination, the drone antenna orientation is updated and the received power is calculated using the `sigstrength` function, with the propagation model defined in the `pm` variable:

```
for az = az_coarse
    for el = el_coarse

        DroneTx.AntennaAngle = [az, el];

        Pr = sigstrength(sites(i), DroneTx, pm);

        if Pr > Pr_best
            Pr_best = Pr;
            az_best = az;
            el_best = el;
        end

    end
end
```

The best received power value is stored in the `Pr_best` variable, while the corresponding angles are saved in `az_best` and `el_best`. In this way, the drone does not simply point the antenna along the geometric line joining it to the receiver, but searches for the orientation that actually provides the best radio link in the simulated scenario.

After this first phase, a second search with a smaller step is performed around the best orientation previously identified:

```
az_fine = az_best + (-2:0.25:2);  
el_fine = el_best + (-2:0.25:2);
```

This fine search improves pointing precision, avoiding the direct use of the coarse scan result. Also in this case, for each pair of angles, the received power is calculated and only the values producing the maximum signal level are retained:

```
for az = az_fine  
    for el = el_fine  
  
        DroneTx.AntennaAngle = [az, el];  
  
        Pr = sigstrength(sites(i), DroneTx, pm);  
  
        if Pr > Pr_best  
            Pr_best = Pr;  
            az_best = az;  
            el_best = el;  
        end  
    end  
end
```

Once the search is completed, the drone directional antenna is oriented according to the pair of angles that produced the maximum received power:

```
DroneTx.AntennaAngle = [az_best, el_best];  
  
az_definitivo = DroneTx.AntennaAngle(1);  
el_definitivo = DroneTx.AntennaAngle(2);
```

The obtained angles are stored as final values, so that they can also be reused in the following scenarios, when the drone is already deployed and may need to update its position.

In summary, in this first scenario the drone is activated when the received power drops below the threshold of -65 dBm. After deployment, the system defines the drone as a new transmitting node, calculates an initial pointing direction toward the team on

foot, and optimizes the orientation of the Yagi-Uda antenna through a two-step angular search. This procedure makes it possible to identify the most favorable pointing in terms of received power, taking into account not only the geometric position of the receiver, but also the effects introduced by the propagation model used in the simulation.

3.3.2 Insufficient coverage with the drone already deployed

The second considered scenario concerns the case in which the received power is still below the threshold of -65 dBm, but the drone has already been deployed in a previous step. In this condition, it is therefore not necessary to initialize the drone again from scratch, but its position and antenna pointing must be updated, searching for a more favorable configuration to maintain the radio link with the team on foot.

In the code, this situation is handled in the `else` branch, which is executed when the drone is already active:

```
else

    if drone_attivo == true

        center_lat = CoordinateDrone(1);
        center_lon = CoordinateDrone(2);
```

The current coordinates of the drone are used as the center of the search area. Starting from this position, the system generates a set of possible new candidate positions, limited by the maximum distance that the drone can travel in one time step. This distance is represented by the `copertura_drone` variable.

To build the search area, a local grid is defined in the East-North-Up coordinate system, that is, a system referred to the current position of the drone. The coordinates along the x and y axes are generated using the `linspace` function:

```
N_punti = 4;

asseX = linspace(-copertura_drone, copertura_drone, N_punti);
asseY = linspace(-copertura_drone, copertura_drone, N_punti);

[Xgrid, Ygrid] = meshgrid(asseX, asseY);
```

In particular, `asseX` and `asseY` contain the possible drone displacements with respect to its current position, respectively along the East-West and North-South directions. The `meshgrid` function combines these values by generating a two-dimensional

grid of candidate points. Each pair (X_{grid}, Y_{grid}) therefore represents a possible new drone position expressed as a local displacement with respect to the central position.

Since vertical displacement is not considered at this stage, the z coordinate is set to zero for all grid points:

```
Zgrid = zeros(size(Xgrid));
```

Subsequently, the points of the local grid are converted into geographic coordinates, namely latitude and longitude, using the `enu2geodetic` function [14]:

```
wgs84 = referenceEllipsoid('wgs84');
```

```
[lat_grid, lon_grid, alt] = enu2geodetic( ...  
    Xgrid, Ygrid, Zgrid, ...  
    center_lat, center_lon, 50, wgs84);
```

This conversion is necessary because MATLAB `txsite` and `rxsite` objects work with geographic coordinates. In this way, the grid built locally around the drone is transformed into a set of possible real positions expressed in latitude and longitude.

The obtained matrices are then converted into vectors, so that each candidate point can be analyzed through a loop:

```
lat_list = lat_grid(:);  
lon_list = lon_grid(:);
```

```
x_list = Xgrid(:);  
y_list = Ygrid(:);
```

```
infoTx = [];
```

For each candidate point, a receiver associated with the drone, indicated as `droneRx`, is temporarily created. This receiver is used to verify whether, in the hypothesized new position, the drone is still able to correctly receive the signal from the PMA:

```
droneRx = rxsite( ...  
    Name="DroneRX", ...  
    Latitude=latD, ...  
    Longitude=lonD, ...  
    AntennaHeight=50, ...  
    Antenna=antenna, ...
```

```
AntennaAngle=[0; -90]);
```

```
ss_in = sigstrength(droneRx, tx, pm);
```

The quantity `ss_in` therefore represents the input power received by the drone. This check is important because the drone must not only improve the link toward the team on foot, but also maintain a valid link with the source node. Otherwise, the new position would not be useful as a relay node.

To prevent the drone from being moved beyond the distance that can be traveled in the considered step, the distance of the candidate point from the center of the grid is calculated:

```
distanza_punto = sqrt(xD^2 + yD^2);
```

The point is considered valid only if it belongs to the area reachable by the drone and if the received input power is greater than the minimum threshold:

```
if distanza_punto <= copertura_drone && ss_in > -65
```

In this way, even though the grid generated with `meshgrid` has a square shape, only the points inside the maximum drone displacement radius are actually accepted. The condition on the distance therefore restricts the search to a circular area compatible with the real motion of the drone in the analyzed time step.

For each valid position, a new `DroneTx` transmitter is created and placed at the candidate coordinates:

```
DroneTx = txsite( ...  
    Name="DroneTX", ...  
    Latitude=latD, ...  
    Longitude=lonD, ...  
    AntennaHeight=50, ...  
    TransmitterFrequency=160e6, ...  
    TransmitterPower=2, ...  
    Antenna=antennaYagi);
```

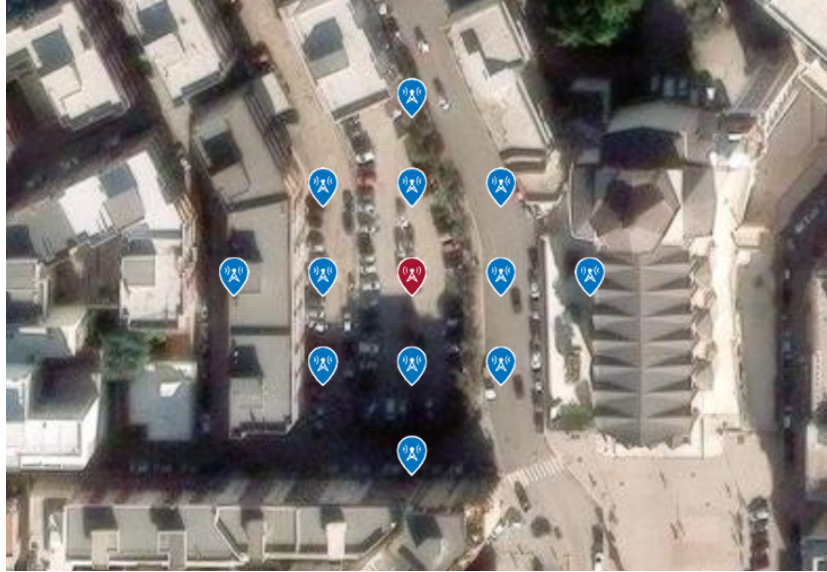


Figure 3.2: Valid candidate positions for drone repositioning within the coverage area.

Figure 3.2 shows the candidate drone positions obtained using the `Meshgrid` function. At this point, for the candidate position, a new optimization of the directional antenna pointing is performed. Unlike the first scenario, in which the drone had just been deployed and the angular search was wider, in this case the process starts from the final angles calculated in the previous step. The search is therefore carried out in a narrower neighborhood:

```
Pr_best = -inf;
az_best = az_definitivo;
el_best = el_definitivo;

az_fine = az_definitivo + (-10:1:10);
el_fine = el_definitivo + (-1:0.25:1);
```

For each pair of angles, the pointing of the drone antenna is updated and the power received by the team on foot is calculated:

```
for az = az_fine
    for el = el_fine

        DroneTx.AntennaAngle = [az, el];

        Pr = sigstrength(sites(i), DroneTx, pm);
```

```

        if Pr > Pr_best
            Pr_best = Pr;
            az_best = az;
            el_best = el;
        end

    end

end
end

```

The result of the evaluation is saved in the `infoTx` matrix, which contains, for each valid point, the candidate coordinates, the best pointing angles, the output power received by the team, and the input power received from the PMA:

```
infoTx = [infoTx; latD, lonD, az_best, el_best, Pr_best, ss_in];
```

After the analysis of all candidate points is completed, the configuration that maximizes the power received by the team on foot is selected:

```
[Pr_max, idx_best] = max(infoTx(:,5));
```

```
bestRow = infoTx(idx_best,:);
```

The fifth column of `infoTx` contains the received power value `Pr_best` associated with each candidate position. The index `idx_best` therefore identifies the row corresponding to the best configuration found.

The drone coordinates and the final antenna angles are updated using the values contained in the best row:

```
CoordinateDrone = [bestRow(1), bestRow(2)];
```

```
az_definitivo = bestRow(3);
```

```
el_definitivo = bestRow(4);
```

```
InPower = bestRow(5);
```

Finally, the drone transmitting site is recreated at the new selected position and the optimal antenna pointing is assigned:

```
DroneTx = txsite( ...
    Name="DroneTX", ...

```



```

Latitude=CoordinateDrone(1), ...
Longitude=CoordinateDrone(2), ...
AntennaHeight=50, ...
TransmitterFrequency=160e6, ...
TransmitterPower=2, ...
Antenna=antennaYagi);

```

```
DroneTx.AntennaAngle = [az_definitivo, el_definitivo];
```

In summary, in this second scenario the drone, already active, is not simply kept at its previous position, but is dynamically repositioned. The procedure generates a grid of possible displacements around the current position through `meshgrid`, converts these points into geographic coordinates, and selects only those compatible with the movement radius and with sufficient received input power. For each valid position, the Yagi-Uda antenna pointing is then optimized, and the configuration that maximizes the power received by the team on foot is chosen. In this way, the drone acts as a mobile support node, able to progressively update its position and antenna orientation according to the propagation conditions present in the simulated urban scenario.

3.3.3 Coverage restoration and drone return

The last considered scenario concerns the case in which the power received by the team on foot returns above the predefined minimum threshold, while the drone is still active. In this condition, the direct link is again considered sufficient and, consequently, the system can evaluate the progressive return of the drone to its base.

In the code, this situation is identified through the following check:

```

if ss > -65 && drone_attivo == true

    k = k + 1;

```

The condition requires the received signal to be greater than -65 dBm and the drone to be actually active. When both conditions are satisfied, the counter k is incremented. This counter was introduced to prevent the drone from returning immediately after the first favorable signal reading. In this way, the system waits for the coverage to remain sufficient for several consecutive steps, reducing the risk of premature returns due to momentary fluctuations in the received power.

The return is started only when the counter reaches a value of at least 4:

```

if k >= 4
    drone_rientro = true;

```

When this condition is satisfied, the `drone_rientro` variable is set to `true`, indicating that the drone no longer has to search for a new coverage position, but must begin moving toward its base. At this point, the base coordinates and the current drone coordinates are defined:

```
lat_base = baseDrone(1);  
lon_base = baseDrone(2);  
  
lat_attuale = CoordinateDrone(1);  
lon_attuale = CoordinateDrone(2);
```

To calculate the distance between the current drone position and the base, the geographic coordinates are converted into local ENU coordinates with respect to the current drone position [15]:

```
[x_base, y_base, z_base] = geodetic2enu(lat_base, lon_base, 0, ...  
    lat_attuale, lon_attuale, 0, wgs84);  
  
distanza_base = sqrt(x_base^2 + y_base^2);
```

The `geodetic2enu` function makes it possible to express the base position in a local system centered on the current drone position. In this system, x represents the eastward displacement and y the northward displacement. The planar distance from the base is then calculated through the Euclidean relationship:

$$d_{\text{base}} = \sqrt{x_{\text{base}}^2 + y_{\text{base}}^2}$$

where d_{base} represents the distance between the drone and the base in the horizontal plane.

At this point, the code verifies whether the base lies within the maximum distance that can be traveled by the drone in the current step, represented by the `copertura_drone` variable:

```
if distanza_base <= copertura_drone  
  
    CoordinateDrone = baseDrone;  
  
    drone_attivo = false;  
    drone_rientro = false;  
    k = 0;
```

If the distance from the base is less than or equal to the maximum allowed displacement, the drone is moved directly back to the base position. In this case, the return can be considered completed: the drone is deactivated, the `drone_rientro` variable is set back to `false`, and the counter k is reset. The system therefore returns to the initial condition, waiting for a possible new signal degradation.

If the base is farther away than the distance that can be traveled in a single step, the drone cannot return immediately. The movement direction toward the base is then calculated through the unit vector associated with the vector connecting the current drone position to the base:

```
versore_x = x_base / distanza_base;
versore_y = y_base / distanza_base;
```

The unit vector identifies the return direction while maintaining unit magnitude. Subsequently, this direction is multiplied by the maximum distance that can be traveled in the step, so as to obtain the actual drone displacement:

```
x_step = versore_x * copertura_drone;
y_step = versore_y * copertura_drone;
```

In this way, the drone moves toward the base by the maximum distance allowed in the current step, without exceeding the limit imposed by its speed and by the simulation time interval.

The new local coordinates are then converted back into latitude and longitude using the `enu2geodetic` function:

```
[lat_new, lon_new, alt_new] = enu2geodetic(x_step, y_step, 0, ...
    lat_attuale, lon_attuale, 0, wgs84);
```

```
CoordinateDrone = [lat_new, lon_new];
```

The new drone position is then updated and will be used in the next simulation step. If, in the following step, the signal also remains above the threshold and the drone has not yet reached the base, the same procedure will be repeated, producing a gradual return.

In summary, this third scenario introduces a controlled return logic for the drone. The return is not started immediately when the signal goes back above the threshold, but only after some consecutive favorable steps, in order to ensure greater decision stability. Once the return has started, the drone moves progressively toward the base: if the base is reachable in the current step, the drone is deactivated; otherwise, a displacement along

the drone-base direction is calculated and limited by the maximum distance that can be traveled in the simulation step. Figure 3.3 shows the movement of the drone along the entire route.

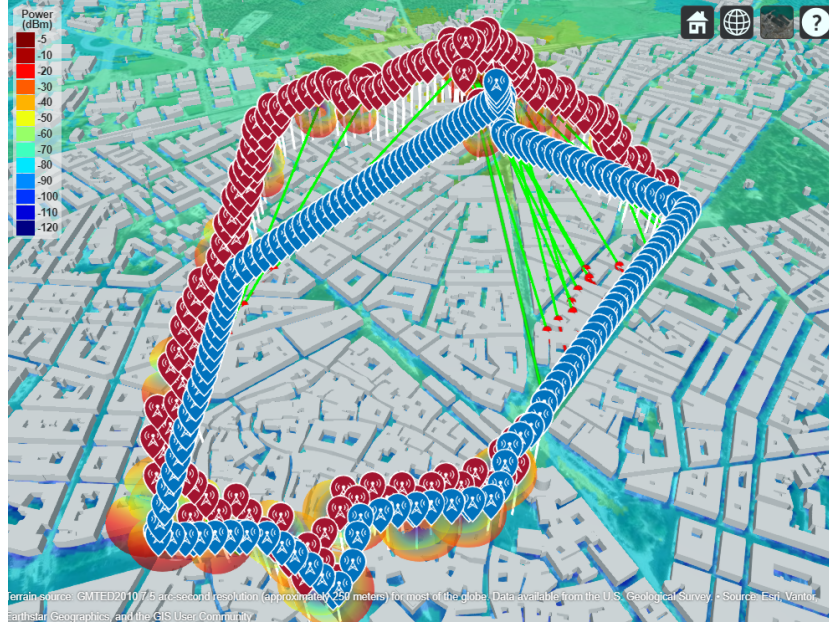


Figure 3.3: Drone positioning along the entire route.

3.4 Performance analysis

The performance analysis aims to quantitatively evaluate the effectiveness of the drone-based support system along the entire procession route. Unlike the previous sections, in which the operating logic of the drone relay was described, this phase focuses on the results obtained from the simulation and on the comparison between the two considered configurations: direct link between the PMA and the team on foot, and drone-assisted link in the critical sections.

For each position of the mobile receiver along the discretized route, the received power was calculated in the case without drone. This quantity represents the main reference for identifying the zones where the direct radio link is sufficient and those where, instead, the signal falls below the operating threshold fixed at -65 dBm. At the points where this condition occurs, the drone is used as a support node and the new power received by the team on foot is evaluated.

The results were organized in a table, indicated in the code with the name `RisultatiFinali`, containing for each point of the route the receiver coordinates, the received power without drone, the received power with drone, the position assumed by the drone, and the pointing angles of the directional antenna. This table is automatically saved in an Excel file. In this way, it was possible to reconstruct both the signal trend along the route and

the actual contribution provided by the drone in the zones with lower coverage.

To make the comparison more immediate, Figure 3.4 shows the trend of received power as a function of the receiver index. The blue curve represents the received power in the direct link, while the orange curve represents the received power considering drone support. In the sections where the drone is not active, the orange curve coincides with the blue one, since the direct link is already sufficient and there is no need to use the relay node. In the critical sections, however, the orange curve separates from the blue one and shows the improvement obtained thanks to the drone intervention.

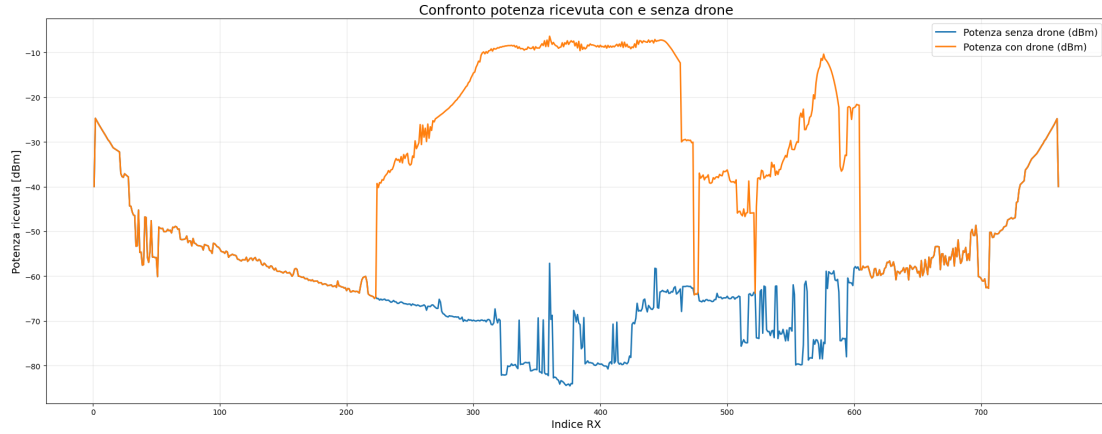


Figure 3.4: Comparison between the received power without drone and the received power with drone-relay support along the procession route.

From Figure 3.4, it can be observed that, in the first part of the route, the received power without drone remains above the activation threshold. In this zone, the direct link between the PMA and the mobile receiver is therefore adequate and the drone does not intervene. Proceeding along the route, however, the received power progressively decreases until it reaches values lower than the predefined threshold. This behavior highlights the presence of radio shadow zones, due to the greater signal attenuation and to the unfavorable propagation conditions present in the urban scenario.

The first drone intervention occurs at RX index 224, where the received power in the direct link assumes a value of approximately -65.04 dBm. At this point, the system detects that the critical threshold has been exceeded and activates the drone relay. With drone support, the received power increases to approximately -39.30 dBm, with an increase equal to:

$$\Delta P = P_{\text{with drone}} - P_{\text{without drone}}$$

$$\Delta P = -39.30 - (-65.04) \simeq 25.74 \text{ dB}$$

This first result shows how the use of the drone makes it possible to recover a significant margin already at the point where the direct link begins to become critical. The improvement is not limited to a single position, but also continues in the following sections, where the orange curve remains clearly above the blue curve.

The most significant behavior is observed in the central zone of the route, where the received power without drone reaches the lowest values of the entire simulation. In particular, the minimum value of the direct link is approximately -84.54 dBm. Without a support node, this power level would represent a strongly penalizing condition for the continuity of radio communication. With the drone intervention, however, the received power in the same section is brought back to values on the order of -7.75 dBm.

The improvement in the worst case can therefore be estimated as:

$$\Delta P = -7.75 - (-84.54) \simeq 76.79 \text{ dB}$$

An increase of this magnitude confirms the effectiveness of the drone relay in compensating for the degradation of the direct link. The advantage derives from the different geometry of the radio link: the drone operates at a higher altitude than the ground receiver and can therefore reduce the shielding effect of buildings. In addition, the use of a directional antenna makes it possible to concentrate the radiated energy toward the team on foot, further increasing the power level available at the receiver.

Table 3.1 reports some representative points of the route, selected among those where the direct link presents the most unfavorable conditions. For each point, the received power without drone, the received power with drone, and the corresponding increase are indicated.

Table 3.1: Comparison between direct link and drone-assisted link at critical points.

RX	$P_{\text{without drone}}$ [dBm]	$P_{\text{with drone}}$ [dBm]	ΔP [dB]
224	-65.04	-39.30	25.74
322	-82.14	-8.83	73.31
356	-81.61	-7.40	74.21
376	-84.54	-7.75	76.79

The reported values confirm that the contribution of the drone becomes particularly relevant at the points where the direct link is most degraded. In the worst cases, the increase in received power exceeds 70 dB, highlighting a very marked variation in reception conditions. This result is consistent with the trend shown in Figure 3.4, where the separation between the two curves increases precisely in the zones with lower coverage.

From the point of view of communication quality, the increase in received power also leads to an improvement in the signal-to-noise ratio. For the same noise level and modulation scheme, greater power at the receiver reduces the bit error probability and makes the link more robust. Consequently, in the sections where the direct signal has very low values, the drone contributes not only to increasing the received power level, but also to making communication between the PMA and the team on foot more reliable.

Another positive aspect concerns the selectivity of the intervention. The drone is not kept continuously active, but operates only when the received power drops below the set threshold. This choice avoids continuous use of the relay node, reducing energy consumption and limiting the operation of the support system to only the truly critical zones. The continuous orange curve shown in Figure 3.4 clearly visualizes this behavior: in non-critical sections the system coincides with the direct link, while in degraded zones the drone significantly modifies the received power level.

In summary, the performance analysis shows that the proposed system is able to significantly improve radio coverage along the procession route. The direct link presents some zones characterized by strong attenuation, with minimum values close to -84.54 dBm. The intervention of the drone relay makes it possible to raise the received power at the critical points, reaching an improvement of approximately 76.79 dB in the worst case. These results confirm the validity of the proposed strategy and show how a mobile radio node can help ensure greater service continuity in complex urban scenarios.

CONCLUSIONS

This work analyzed the radio coverage along the route of the Festa dei Santi Medici in Bitonto, with the objective of evaluating the quality of the link between the Advanced Medical Post and a mobile team engaged during the procession. The simulation was developed in MATLAB, using the Longley-Rice propagation model and a discretized representation of the real route.

The obtained results highlighted the presence of some critical sections, where the received power of the direct link falls below the reference threshold equal to -65 dBm. In particular, in the case without drone, a minimum value of approximately -84.54 dBm was detected, confirming that, in an urban scenario, direct communication alone may not always guarantee an adequate signal level.

To improve link continuity, a drone-relay support strategy was therefore proposed. The drone intervenes only at the points where the direct signal is insufficient, searches for a favorable position, and orients the directional antenna toward the team on foot. The performance analysis showed a significant increase in received power in the critical sections: at the first activation point the improvement is approximately 25.74 dB, while in the worst case it reaches approximately 76.79 dB.

These results confirm the effectiveness of the proposed solution, showing how a mobile radio node can help reduce shadow zones and improve the reliability of operational communications during complex public events. A relevant aspect is also the selectivity of the intervention: the drone is not kept continuously active, but is used only when necessary, making the strategy more compatible with autonomy constraints and real operating conditions.

The developed model nevertheless represents a first approximation of the operational scenario. In future developments, it would be appropriate to use more detailed propagation models for urban environments, such as ray tracing, and to integrate additional link quality parameters, such as signal-to-noise ratio, E_b/N_0 , and bit error rate. Overall, the project shows how the use of a drone relay can represent a promising solution for increasing the continuity and reliability of radio communications in dynamic urban scenarios.

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APPENDIX

Tabella completa dei risultati della simulazione

In questa appendice viene riportata la tabella completa dei risultati ottenuti dalla simulazione. Per ciascun punto del percorso sono indicati l'indice del ricevitore, le coordinate geografiche del ricevitore, la potenza ricevuta in assenza del drone, la potenza ricevuta con l'ausilio del drone-relay, la posizione assunta dal drone e gli angoli di orientamento dell'antenna direttiva Yagi.

Table 3.2: Tabella completa dei risultati della simulazione

RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
1	41.114189	16.691481	-39.9803	NaN	NaN	NaN	NaN	NaN
2	41.114159	16.691481	-24.7348	NaN	NaN	NaN	NaN	NaN
3	41.114129	16.691481	-25.2436	NaN	NaN	NaN	NaN	NaN
4	41.114099	16.691481	-25.7621	NaN	NaN	NaN	NaN	NaN
5	41.114069	16.691481	-26.2448	NaN	NaN	NaN	NaN	NaN
6	41.114039	16.691481	-26.7174	NaN	NaN	NaN	NaN	NaN
7	41.114009	16.691481	-27.1932	NaN	NaN	NaN	NaN	NaN
8	41.113979	16.691481	-27.6508	NaN	NaN	NaN	NaN	NaN
9	41.113949	16.691481	-28.1049	NaN	NaN	NaN	NaN	NaN
10	41.113919	16.691481	-28.5776	NaN	NaN	NaN	NaN	NaN
11	41.11389	16.691481	-29.081	NaN	NaN	NaN	NaN	NaN
12	41.11386	16.691481	-29.5667	NaN	NaN	NaN	NaN	NaN
13	41.11383	16.691481	-29.871	NaN	NaN	NaN	NaN	NaN
14	41.1138	16.691481	-30.3469	NaN	NaN	NaN	NaN	NaN
15	41.11377	16.691481	-30.8324	NaN	NaN	NaN	NaN	NaN
16	41.11374	16.691481	-31.307	NaN	NaN	NaN	NaN	NaN
17	41.11372	16.691459	-31.4937	NaN	NaN	NaN	NaN	NaN
18	41.1137	16.691437	-31.6707	NaN	NaN	NaN	NaN	NaN
19	41.113679	16.691415	-31.8581	NaN	NaN	NaN	NaN	NaN
20	41.113659	16.691393	-32.0555	NaN	NaN	NaN	NaN	NaN
21	41.113639	16.69137	-32.2616	NaN	NaN	NaN	NaN	NaN
22	41.113619	16.691348	-37.0236	NaN	NaN	NaN	NaN	NaN
23	41.113599	16.691326	-37.759	NaN	NaN	NaN	NaN	NaN
24	41.113578	16.691304	-37.9297	NaN	NaN	NaN	NaN	NaN
25	41.113558	16.691281	-37.1261	NaN	NaN	NaN	NaN	NaN
26	41.113538	16.691259	-37.3682	NaN	NaN	NaN	NaN	NaN
27	41.113518	16.691237	-37.6147	NaN	NaN	NaN	NaN	NaN
28	41.113498	16.691215	-37.8505	NaN	NaN	NaN	NaN	NaN
29	41.113477	16.691193	-44.3229	NaN	NaN	NaN	NaN	NaN
30	41.113457	16.69117	-44.4025	NaN	NaN	NaN	NaN	NaN
31	41.113438	16.691148	-45.6326	NaN	NaN	NaN	NaN	NaN
32	41.113418	16.691125	-46.3452	NaN	NaN	NaN	NaN	NaN
33	41.113399	16.691103	-46.4525	NaN	NaN	NaN	NaN	NaN
34	41.11338	16.69108	-53.3119	NaN	NaN	NaN	NaN	NaN
35	41.11336	16.691057	-53.2535	NaN	NaN	NaN	NaN	NaN
36	41.113341	16.691035	-45.2457	NaN	NaN	NaN	NaN	NaN
37	41.113322	16.691012	-54.7073	NaN	NaN	NaN	NaN	NaN
38	41.113302	16.69099	-54.6291	NaN	NaN	NaN	NaN	NaN
39	41.113283	16.690967	-57.5584	NaN	NaN	NaN	NaN	NaN
40	41.113264	16.690944	-57.4332	NaN	NaN	NaN	NaN	NaN
41	41.113244	16.690922	-46.774	NaN	NaN	NaN	NaN	NaN
42	41.113225	16.690899	-46.8942	NaN	NaN	NaN	NaN	NaN
43	41.113205	16.690877	-55.8053	NaN	NaN	NaN	NaN	NaN
44	41.113186	16.690854	-56.9238	NaN	NaN	NaN	NaN	NaN
45	41.113167	16.690832	-54.575	NaN	NaN	NaN	NaN	NaN
46	41.113147	16.690809	-47.6255	NaN	NaN	NaN	NaN	NaN
47	41.113128	16.690786	-55.7463	NaN	NaN	NaN	NaN	NaN
48	41.113109	16.690764	-55.7247	NaN	NaN	NaN	NaN	NaN

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
49	41.113089	16.690741	-55.8716	NaN	NaN	NaN	NaN	NaN
50	41.11307	16.69072	-55.8471	NaN	NaN	NaN	NaN	NaN
51	41.11305	16.690699	-60.1092	NaN	NaN	NaN	NaN	NaN
52	41.113031	16.690678	-49.0303	NaN	NaN	NaN	NaN	NaN
53	41.113011	16.690657	-49.2938	NaN	NaN	NaN	NaN	NaN
54	41.112992	16.690636	-49.4053	NaN	NaN	NaN	NaN	NaN
55	41.112972	16.690615	-49.3082	NaN	NaN	NaN	NaN	NaN
56	41.112953	16.690594	-50.0694	NaN	NaN	NaN	NaN	NaN
57	41.112934	16.690573	-50.0577	NaN	NaN	NaN	NaN	NaN
58	41.112914	16.690552	-50.0002	NaN	NaN	NaN	NaN	NaN
59	41.112895	16.690532	-49.5408	NaN	NaN	NaN	NaN	NaN
60	41.112875	16.690511	-49.8716	NaN	NaN	NaN	NaN	NaN
61	41.112856	16.69049	-49.7349	NaN	NaN	NaN	NaN	NaN
62	41.112836	16.690469	-50.3556	NaN	NaN	NaN	NaN	NaN
63	41.112817	16.690448	-49.0592	NaN	NaN	NaN	NaN	NaN
64	41.112797	16.690427	-49.1505	NaN	NaN	NaN	NaN	NaN
65	41.112778	16.690406	-48.8877	NaN	NaN	NaN	NaN	NaN
66	41.112758	16.690385	-48.9888	NaN	NaN	NaN	NaN	NaN
67	41.112739	16.690364	-49.5424	NaN	NaN	NaN	NaN	NaN
68	41.112719	16.690343	-49.4743	NaN	NaN	NaN	NaN	NaN
69	41.1127	16.690322	-51.7595	NaN	NaN	NaN	NaN	NaN
70	41.11268	16.690301	-51.8544	NaN	NaN	NaN	NaN	NaN
71	41.112661	16.69028	-51.8141	NaN	NaN	NaN	NaN	NaN
72	41.112641	16.690259	-51.7865	NaN	NaN	NaN	NaN	NaN
73	41.112622	16.690238	-51.6463	NaN	NaN	NaN	NaN	NaN
74	41.112602	16.690217	-51.0528	NaN	NaN	NaN	NaN	NaN
75	41.112583	16.690196	-52.5175	NaN	NaN	NaN	NaN	NaN
76	41.112564	16.690175	-52.1263	NaN	NaN	NaN	NaN	NaN
77	41.112544	16.690154	-52.1789	NaN	NaN	NaN	NaN	NaN
78	41.112525	16.690133	-53.2565	NaN	NaN	NaN	NaN	NaN
79	41.112505	16.690112	-51.5896	NaN	NaN	NaN	NaN	NaN
80	41.112486	16.690091	-52.6299	NaN	NaN	NaN	NaN	NaN
81	41.112466	16.69007	-52.7248	NaN	NaN	NaN	NaN	NaN
82	41.112447	16.690049	-53.0508	NaN	NaN	NaN	NaN	NaN
83	41.112428	16.690028	-53.1989	NaN	NaN	NaN	NaN	NaN
84	41.112409	16.690007	-53.2119	NaN	NaN	NaN	NaN	NaN
85	41.11239	16.689986	-53.3057	NaN	NaN	NaN	NaN	NaN
86	41.112371	16.689965	-53.4006	NaN	NaN	NaN	NaN	NaN
87	41.112352	16.689944	-54.1695	NaN	NaN	NaN	NaN	NaN
88	41.112333	16.689923	-52.9386	NaN	NaN	NaN	NaN	NaN
89	41.112314	16.689902	-53.0513	NaN	NaN	NaN	NaN	NaN
90	41.112295	16.689881	-53.1611	NaN	NaN	NaN	NaN	NaN
91	41.112276	16.68986	-53.2946	NaN	NaN	NaN	NaN	NaN
92	41.112257	16.68984	-54.2541	NaN	NaN	NaN	NaN	NaN
93	41.112238	16.689819	-54.337	NaN	NaN	NaN	NaN	NaN
94	41.112219	16.689798	-54.9323	NaN	NaN	NaN	NaN	NaN
95	41.1122	16.689777	-52.6612	NaN	NaN	NaN	NaN	NaN
96	41.112181	16.689756	-52.7567	NaN	NaN	NaN	NaN	NaN
97	41.112162	16.689735	-53.0715	NaN	NaN	NaN	NaN	NaN
98	41.112143	16.689714	-53.4257	NaN	NaN	NaN	NaN	NaN
99	41.112124	16.689693	-53.544	NaN	NaN	NaN	NaN	NaN
100	41.112105	16.689672	-53.8705	NaN	NaN	NaN	NaN	NaN
101	41.112086	16.689651	-54.4157	NaN	NaN	NaN	NaN	NaN
102	41.112067	16.68963	-54.5931	NaN	NaN	NaN	NaN	NaN
103	41.112048	16.689609	-54.6117	NaN	NaN	NaN	NaN	NaN
104	41.112029	16.689588	-54.9153	NaN	NaN	NaN	NaN	NaN
105	41.11201	16.689567	-54.4214	NaN	NaN	NaN	NaN	NaN
106	41.111991	16.689546	-54.5997	NaN	NaN	NaN	NaN	NaN
107	41.111972	16.689525	-55.7977	NaN	NaN	NaN	NaN	NaN
108	41.111953	16.689504	-55.4174	NaN	NaN	NaN	NaN	NaN
109	41.111934	16.689483	-55.3106	NaN	NaN	NaN	NaN	NaN
110	41.111915	16.689462	-55.1163	NaN	NaN	NaN	NaN	NaN
111	41.111896	16.689441	-55.3037	NaN	NaN	NaN	NaN	NaN
112	41.111877	16.68942	-55.3296	NaN	NaN	NaN	NaN	NaN
113	41.111858	16.689399	-55.4829	NaN	NaN	NaN	NaN	NaN
114	41.111839	16.689378	-56.2041	NaN	NaN	NaN	NaN	NaN
115	41.11182	16.689357	-56.2798	NaN	NaN	NaN	NaN	NaN
116	41.111801	16.689336	-56.4758	NaN	NaN	NaN	NaN	NaN
117	41.111782	16.689315	-56.6127	NaN	NaN	NaN	NaN	NaN
118	41.111763	16.689293	-56.3339	NaN	NaN	NaN	NaN	NaN
119	41.111744	16.689272	-56.3532	NaN	NaN	NaN	NaN	NaN
120	41.111725	16.689251	-56.3943	NaN	NaN	NaN	NaN	NaN

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
121	41.111706	16.689229	-55.7958	NaN	NaN	NaN	NaN	NaN
122	41.111687	16.689208	-56.7052	NaN	NaN	NaN	NaN	NaN
123	41.111668	16.689187	-56.4106	NaN	NaN	NaN	NaN	NaN
124	41.111649	16.689165	-56.0043	NaN	NaN	NaN	NaN	NaN
125	41.11163	16.689144	-55.9751	NaN	NaN	NaN	NaN	NaN
126	41.111611	16.689123	-56.4814	NaN	NaN	NaN	NaN	NaN
127	41.111592	16.689101	-56.0685	NaN	NaN	NaN	NaN	NaN
128	41.111573	16.68908	-55.8416	NaN	NaN	NaN	NaN	NaN
129	41.111554	16.689059	-56.4783	NaN	NaN	NaN	NaN	NaN
130	41.111535	16.689037	-56.575	NaN	NaN	NaN	NaN	NaN
131	41.111516	16.689016	-56.2631	NaN	NaN	NaN	NaN	NaN
132	41.111497	16.688995	-56.5234	NaN	NaN	NaN	NaN	NaN
133	41.111478	16.688974	-57.3521	NaN	NaN	NaN	NaN	NaN
134	41.111459	16.688952	-57.5742	NaN	NaN	NaN	NaN	NaN
135	41.11144	16.688931	-57.1205	NaN	NaN	NaN	NaN	NaN
136	41.111421	16.68891	-57.3761	NaN	NaN	NaN	NaN	NaN
137	41.111402	16.688888	-57.8055	NaN	NaN	NaN	NaN	NaN
138	41.111383	16.688867	-57.8588	NaN	NaN	NaN	NaN	NaN
139	41.111364	16.688846	-58.1995	NaN	NaN	NaN	NaN	NaN
140	41.111345	16.688824	-57.7467	NaN	NaN	NaN	NaN	NaN
141	41.111326	16.688803	-57.7829	NaN	NaN	NaN	NaN	NaN
142	41.111307	16.688782	-58.2857	NaN	NaN	NaN	NaN	NaN
143	41.111288	16.68876	-58.6287	NaN	NaN	NaN	NaN	NaN
144	41.111269	16.688739	-58.2321	NaN	NaN	NaN	NaN	NaN
145	41.11125	16.688718	-58.8593	NaN	NaN	NaN	NaN	NaN
146	41.111231	16.688696	-58.9608	NaN	NaN	NaN	NaN	NaN
147	41.111212	16.688675	-58.8936	NaN	NaN	NaN	NaN	NaN
148	41.111193	16.688654	-59.0219	NaN	NaN	NaN	NaN	NaN
149	41.111174	16.688633	-59.1277	NaN	NaN	NaN	NaN	NaN
150	41.111155	16.688611	-59.306	NaN	NaN	NaN	NaN	NaN
151	41.111136	16.68859	-59.0422	NaN	NaN	NaN	NaN	NaN
152	41.111117	16.688569	-58.9362	NaN	NaN	NaN	NaN	NaN
153	41.111098	16.688547	-59.2884	NaN	NaN	NaN	NaN	NaN
154	41.111079	16.688526	-59.3143	NaN	NaN	NaN	NaN	NaN
155	41.11106	16.688505	-59.1853	NaN	NaN	NaN	NaN	NaN
156	41.111042	16.688484	-59.5451	NaN	NaN	NaN	NaN	NaN
157	41.111023	16.688463	-60.043	NaN	NaN	NaN	NaN	NaN
158	41.111004	16.688442	-59.605	NaN	NaN	NaN	NaN	NaN
159	41.110985	16.68842	-58.4478	NaN	NaN	NaN	NaN	NaN
160	41.110966	16.688399	-58.2922	NaN	NaN	NaN	NaN	NaN
161	41.110947	16.688378	-58.5327	NaN	NaN	NaN	NaN	NaN
162	41.110928	16.688357	-60.0238	NaN	NaN	NaN	NaN	NaN
163	41.11091	16.688336	-60.0442	NaN	NaN	NaN	NaN	NaN
164	41.110891	16.688314	-60.165	NaN	NaN	NaN	NaN	NaN
165	41.110872	16.688293	-60.2947	NaN	NaN	NaN	NaN	NaN
166	41.110853	16.688272	-60.2035	NaN	NaN	NaN	NaN	NaN
167	41.110834	16.688251	-60.5563	NaN	NaN	NaN	NaN	NaN
168	41.110815	16.68823	-60.5797	NaN	NaN	NaN	NaN	NaN
169	41.110797	16.688209	-60.5589	NaN	NaN	NaN	NaN	NaN
170	41.110778	16.688187	-60.9434	NaN	NaN	NaN	NaN	NaN
171	41.110759	16.688166	-60.8542	NaN	NaN	NaN	NaN	NaN
172	41.11074	16.688145	-61.0426	NaN	NaN	NaN	NaN	NaN
173	41.110721	16.688124	-60.8237	NaN	NaN	NaN	NaN	NaN
174	41.110702	16.688103	-61.1645	NaN	NaN	NaN	NaN	NaN
175	41.110683	16.688081	-61.1341	NaN	NaN	NaN	NaN	NaN
176	41.110665	16.68806	-61.1152	NaN	NaN	NaN	NaN	NaN
177	41.110646	16.688039	-61.1158	NaN	NaN	NaN	NaN	NaN
178	41.110627	16.688018	-61.5293	NaN	NaN	NaN	NaN	NaN
179	41.110608	16.687997	-61.4715	NaN	NaN	NaN	NaN	NaN
180	41.110589	16.687976	-61.4638	NaN	NaN	NaN	NaN	NaN
181	41.11057	16.687954	-61.8844	NaN	NaN	NaN	NaN	NaN
182	41.110551	16.687932	-61.7788	NaN	NaN	NaN	NaN	NaN
183	41.110532	16.68791	-61.7421	NaN	NaN	NaN	NaN	NaN
184	41.110513	16.687889	-61.8237	NaN	NaN	NaN	NaN	NaN
185	41.110494	16.687867	-62.1449	NaN	NaN	NaN	NaN	NaN
186	41.110475	16.687845	-62.0702	NaN	NaN	NaN	NaN	NaN
187	41.110456	16.687824	-62.0255	NaN	NaN	NaN	NaN	NaN
188	41.110437	16.687802	-61.9888	NaN	NaN	NaN	NaN	NaN
189	41.110418	16.68778	-62.3454	NaN	NaN	NaN	NaN	NaN
190	41.110399	16.687759	-62.268	NaN	NaN	NaN	NaN	NaN
191	41.11038	16.687737	-62.1632	NaN	NaN	NaN	NaN	NaN
192	41.110361	16.687715	-62.5138	NaN	NaN	NaN	NaN	NaN

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
193	41.110342	16.687694	-61.1011	NaN	NaN	NaN	NaN	NaN
194	41.110323	16.687672	-62.2006	NaN	NaN	NaN	NaN	NaN
195	41.110304	16.68765	-62.2601	NaN	NaN	NaN	NaN	NaN
196	41.110285	16.687629	-62.4178	NaN	NaN	NaN	NaN	NaN
197	41.110266	16.687607	-62.4471	NaN	NaN	NaN	NaN	NaN
198	41.110247	16.687585	-62.6562	NaN	NaN	NaN	NaN	NaN
199	41.110228	16.687564	-62.609	NaN	NaN	NaN	NaN	NaN
200	41.110209	16.687542	-63.2496	NaN	NaN	NaN	NaN	NaN
201	41.11019	16.68752	-62.7633	NaN	NaN	NaN	NaN	NaN
202	41.110171	16.687499	-62.837	NaN	NaN	NaN	NaN	NaN
203	41.110152	16.687477	-63.5283	NaN	NaN	NaN	NaN	NaN
204	41.110132	16.687455	-63.4979	NaN	NaN	NaN	NaN	NaN
205	41.110111	16.687433	-63.2802	NaN	NaN	NaN	NaN	NaN
206	41.110089	16.687411	-63.411	NaN	NaN	NaN	NaN	NaN
207	41.110067	16.687388	-63.4791	NaN	NaN	NaN	NaN	NaN
208	41.110045	16.687366	-63.3884	NaN	NaN	NaN	NaN	NaN
209	41.110023	16.687344	-63.5434	NaN	NaN	NaN	NaN	NaN
210	41.110001	16.687322	-63.79	NaN	NaN	NaN	NaN	NaN
211	41.109969	16.687308	-62.1828	NaN	NaN	NaN	NaN	NaN
212	41.109936	16.687294	-60.8724	NaN	NaN	NaN	NaN	NaN
213	41.109903	16.687279	-60.3226	NaN	NaN	NaN	NaN	NaN
214	41.109874	16.687278	-60.1675	NaN	NaN	NaN	NaN	NaN
215	41.109845	16.687277	-60.0653	NaN	NaN	NaN	NaN	NaN
216	41.109816	16.687275	-61.2589	NaN	NaN	NaN	NaN	NaN
217	41.109787	16.687274	-63.9688	NaN	NaN	NaN	NaN	NaN
218	41.109758	16.687272	-64.2295	NaN	NaN	NaN	NaN	NaN
219	41.109729	16.687271	-64.4062	NaN	NaN	NaN	NaN	NaN
220	41.1097	16.687269	-64.4699	NaN	NaN	NaN	NaN	NaN
221	41.109671	16.687268	-64.6136	NaN	NaN	NaN	NaN	NaN
222	41.109642	16.687266	-64.9966	NaN	NaN	NaN	NaN	NaN
223	41.109613	16.687265	-64.6861	NaN	NaN	NaN	NaN	NaN
224	41.109583	16.687264	-65.0403	-39.3018	41.114503	16.690851	47.0363	24.8935
225	41.109554	16.687262	-65.0342	-40.2111	41.114388	16.6907	48.0363	24.3935
226	41.109525	16.687261	-65.3623	-39.0961	41.114274	16.690548	48.0363	24.8935
227	41.109496	16.687259	-65.1402	-39.2175	41.114159	16.690397	49.0363	24.3935
228	41.109467	16.687258	-65.342	-38.443	41.114045	16.690245	49.0363	25.3935
229	41.109438	16.687256	-65.2269	-38.5558	41.11393	16.690094	50.0363	24.6435
230	41.109409	16.687255	-65.1437	-37.7792	41.113816	16.689943	51.0363	25.6435
231	41.10938	16.687253	-65.4107	-37.4017	41.113701	16.689791	52.0363	26.1435
232	41.109351	16.687252	-65.3915	-37.0649	41.113587	16.68964	53.0363	26.3935
233	41.109322	16.687251	-65.3764	-36.4143	41.113472	16.689488	54.0363	26.6435
234	41.109293	16.687249	-65.6102	-36.829	41.113358	16.68964	52.0363	26.3935
235	41.109264	16.687248	-65.8556	-36.256	41.113243	16.689488	53.0363	26.8935
236	41.109235	16.687246	-65.7308	-36.1546	41.113129	16.689337	54.0363	26.8935
237	41.109206	16.687245	-65.4549	-35.4245	41.113014	16.689185	55.0363	27.1435
238	41.109177	16.687243	-65.7294	-34.3706	41.113129	16.689034	57.0363	26.1435
239	41.109147	16.687243	-65.7551	-33.7261	41.113243	16.688883	59.0363	26.8935
240	41.109118	16.687242	-66.0894	-34.1133	41.113358	16.689034	58.0363	26.3935
241	41.109089	16.687241	-66.1052	-33.9241	41.113243	16.688883	59.0363	26.6435
242	41.10906	16.68724	-65.8773	-34.5307	41.113358	16.689034	58.0363	26.3935
243	41.109031	16.68724	-65.7762	-33.2886	41.113243	16.688883	59.0363	26.3935
244	41.109002	16.687239	-65.9058	-34.7071	41.113358	16.689034	58.0363	26.1435
245	41.108973	16.687238	-66.1396	-32.8382	41.113243	16.688883	60.0363	26.6435
246	41.108944	16.687237	-66.1627	-33.7009	41.113358	16.689034	59.0363	26.3935
247	41.108915	16.687237	-66.2137	-32.9149	41.113243	16.688883	60.0363	26.3935
248	41.108886	16.687236	-66.3123	-32.5684	41.113129	16.688731	61.0363	25.6435
249	41.108856	16.687235	-66.249	-34.7498	41.113243	16.688883	60.0363	26.3935
250	41.108827	16.687234	-66.2494	-35.2118	41.113129	16.688731	61.0363	25.3935
251	41.108798	16.687234	-66.5538	-34.9482	41.113014	16.68858	62.0363	25.6435
252	41.108769	16.687233	-66.233	-33.1616	41.113129	16.688428	64.0363	26.6435
253	41.10874	16.687232	-66.3817	-33.515	41.113243	16.688277	66.0363	26.3935
254	41.108711	16.687231	-66.453	-29.773	41.113129	16.688126	67.0363	26.6435
255	41.108682	16.687231	-66.613	-31.5095	41.113014	16.687974	68.0363	26.6435
256	41.108653	16.68723	-66.4531	-30.9221	41.1129	16.687823	70.0363	26.8935
257	41.108624	16.687229	-66.5714	-30.143	41.112785	16.687671	71.0363	26.8935
258	41.108595	16.687228	-66.7619	-26.1333	41.112671	16.68752	73.0363	26.1435
259	41.108565	16.687228	-66.6681	-30.6022	41.112556	16.687368	74.0363	27.1435
260	41.108536	16.687227	-66.608	-26.2549	41.112671	16.68752	73.0363	26.1435
261	41.108507	16.687226	-66.4312	-28.9451	41.112556	16.687368	74.0363	27.1435
262	41.108478	16.687225	-66.6987	-26.9481	41.112671	16.68752	73.0363	26.1435
263	41.108449	16.687224	-67.6224	-29.9334	41.112556	16.687368	74.0363	27.1435
264	41.108419	16.687222	-66.7592	-26.0213	41.112442	16.68752	73.0363	26.1435

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
265	41.108389	16.68722	-66.7602	-29.185	41.112327	16.687368	74.0363	27.1435
266	41.108359	16.687218	-66.8234	-26.6351	41.112213	16.687217	76.0363	27.6435
267	41.108328	16.687216	-66.4916	-27.4366	41.112098	16.687368	74.0363	26.6435
268	41.108298	16.687213	-66.4369	-25.2789	41.111984	16.687217	76.0363	27.6435
269	41.108268	16.687211	-66.7372	-25.5357	41.111869	16.687368	74.0363	26.8935
270	41.108238	16.687209	-67.0085	-24.8545	41.111755	16.687217	76.0363	27.1435
271	41.108208	16.687207	-67.2244	-24.6521	41.11164	16.687368	74.0363	27.3935
272	41.108178	16.687204	-67.2232	-24.4342	41.111526	16.687217	76.0363	27.6435
273	41.108148	16.687202	-65.1831	-24.2227	41.111411	16.687368	74.0363	27.8935
274	41.108117	16.6872	-65.8254	-23.9929	41.111297	16.687217	75.0363	27.6435
275	41.108087	16.687198	-67.8934	-23.7922	41.111182	16.687368	73.0363	28.6435
276	41.108054	16.687194	-68.3943	-23.5729	41.111068	16.687217	75.0363	29.6435
277	41.108021	16.687191	-68.5444	-23.3133	41.110953	16.687368	73.0363	28.6435
278	41.107988	16.687188	-68.8853	-23.0665	41.110839	16.687217	75.0363	28.8935
279	41.107955	16.687185	-69.0273	-22.8318	41.110724	16.687368	73.0363	29.1435
280	41.107922	16.687182	-69.135	-22.615	41.11061	16.687217	75.0363	30.1435
281	41.107893	16.687181	-68.865	-22.3573	41.110495	16.687368	72.0363	31.1435
282	41.107864	16.68718	-68.9523	-22.0152	41.11038	16.687217	75.0363	30.1435
283	41.107835	16.68718	-68.9552	-21.7953	41.110266	16.687368	72.0363	31.1435
284	41.107806	16.687179	-69.0676	-21.4838	41.110151	16.687217	75.0363	32.1435
285	41.107777	16.687178	-69.137	-21.1303	41.110037	16.687368	72.0363	31.3935
286	41.107748	16.687178	-69.0944	-20.7929	41.109922	16.687217	75.0363	31.8935
287	41.107719	16.687177	-69.1862	-20.5593	41.109808	16.687368	71.0363	32.8935
288	41.10769	16.687177	-69.1137	-20.1146	41.109693	16.687217	75.0363	33.1435
289	41.107661	16.687176	-69.5679	-19.8712	41.109579	16.687368	71.0363	34.1435
290	41.107632	16.687175	-69.549	-19.3825	41.109464	16.687217	74.0363	34.1435
291	41.107604	16.687175	-69.1748	-19.0004	41.10935	16.687066	78.0363	34.8935
292	41.107575	16.687174	-70.1298	-18.5877	41.109235	16.687217	74.0363	35.6435
293	41.107546	16.687174	-69.7729	-18.1726	41.109121	16.687066	78.0363	36.6435
294	41.107516	16.687172	-70.001	-17.723	41.109006	16.687217	74.0363	37.6435
295	41.107487	16.687171	-70.0567	-17.4437	41.108892	16.687066	78.0363	38.6435
296	41.107458	16.68717	-69.7011	-16.7773	41.108777	16.687217	73.0363	39.6435
297	41.107428	16.687169	-69.8531	-16.498	41.108663	16.687066	78.0363	40.6435
298	41.107399	16.687168	-69.8031	-15.7347	41.108548	16.687217	73.0363	41.6435
299	41.10737	16.687167	-70.0081	-15.1876	41.108434	16.687066	78.0363	42.6435
300	41.107341	16.687166	-69.8471	-14.5877	41.108319	16.687217	72.0363	43.6435
301	41.107311	16.687165	-69.9881	-14.4034	41.108205	16.687066	77.0363	44.6435
302	41.107282	16.687164	-70.0668	-13.341	41.10809	16.687217	70.0363	45.6435
303	41.107253	16.687163	-69.9319	-12.7768	41.107976	16.687368	60.0363	46.6435
304	41.107223	16.687161	-69.9	-11.9622	41.107861	16.687217	68.0363	47.6435
305	41.107194	16.68716	-70.0327	-12.1425	41.107747	16.687066	76.0363	48.6435
306	41.107165	16.687159	-69.8154	-10.6334	41.107632	16.687217	66.0363	49.6435
307	41.107135	16.687158	-69.8476	-10.1002	41.107518	16.687066	76.0363	50.6435
308	41.107106	16.687157	-69.983	-9.9006	41.107403	16.687217	66.0363	51.6435
309	41.107077	16.687156	-69.8786	-10.3134	41.107518	16.687066	76.0363	52.6435
310	41.107048	16.687155	-70.2178	-9.8358	41.107403	16.687217	66.0363	53.6435
311	41.107018	16.687154	-69.7614	-10.1589	41.107289	16.687368	56.0363	54.6435
312	41.106989	16.687153	-69.6889	-9.6606	41.107174	16.687217	47.0363	55.6435
313	41.10696	16.687152	-69.9049	-9.599	41.107289	16.687368	40.0363	56.6435
314	41.10693	16.68715	-70.7997	-9.2147	41.107174	16.687217	50.0363	57.6435
315	41.106901	16.687149	-70.8936	-9.7107	41.10706	16.687368	40.0363	58.6435
316	41.1069	16.687178	-70.4556	-9.1239	41.107174	16.687217	50.0363	59.6435
317	41.106899	16.687208	-67.3366	-9.2561	41.10706	16.687368	40.0363	60.6435
318	41.106898	16.687237	-69.3702	-9.1994	41.107174	16.687217	50.0363	61.6435
319	41.106897	16.687266	-70.4526	-8.9551	41.10706	16.687368	40.0363	62.6435
320	41.106897	16.687295	-69.6107	-9.0139	41.107174	16.68752	34.0363	63.6435
321	41.106896	16.687324	-69.8941	-8.8019	41.10706	16.687368	44.0363	64.6435
322	41.106895	16.687353	-82.1378	-8.8255	41.107174	16.68752	40.0363	65.6435
323	41.106894	16.687382	-82.0466	-8.686	41.10706	16.687368	50.0363	66.6435
324	41.106893	16.687411	-82.1025	-8.6698	41.107174	16.68752	48.0363	67.6435
325	41.106892	16.68744	-82.1094	-8.5712	41.10706	16.687368	58.0363	68.6435
326	41.106891	16.687469	-81.9754	-8.5504	41.107174	16.68752	56.0363	69.6435
327	41.10689	16.687499	-80.0736	-8.5002	41.10706	16.687368	66.0363	70.6435
328	41.106889	16.687528	-80.2121	-8.4751	41.107174	16.68752	65.0363	71.6435
329	41.106891	16.68756	-80.0906	-8.4952	41.10706	16.687368	75.0363	72.6435
330	41.106893	16.687592	-79.6195	-8.42	41.107174	16.68752	75.0363	73.6435
331	41.106895	16.687624	-79.9764	-8.5756	41.10706	16.687368	85.0363	74.6435
332	41.106897	16.687656	-79.8851	-8.4412	41.107174	16.68752	84.0363	75.6435
333	41.106899	16.687688	-79.7631	-8.8015	41.10706	16.687368	94.0363	76.6435
334	41.106901	16.68772	-80.3013	-8.5594	41.107174	16.68752	94.0363	77.6435
335	41.106903	16.687752	-80.6742	-9.1061	41.10706	16.687368	104.0363	78.6435
336	41.106905	16.687784	-69.8498	-8.7702	41.107174	16.68752	103.0363	77.6435

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
337	41.106907	16.687817	-79.7574	-9.1896	41.10706	16.687671	93.0363	78.6435
338	41.106908	16.68785	-79.6608	-9.1341	41.107174	16.68752	103.0363	77.6435
339	41.106908	16.687883	-79.7032	-9.1817	41.10706	16.687671	103.0363	78.6435
340	41.106908	16.687916	-79.4231	-9.495	41.107174	16.68752	113.0363	77.6435
341	41.106909	16.687949	-79.2755	-9.2709	41.10706	16.687671	113.0363	78.6435
342	41.106909	16.687982	-79.3591	-9.316	41.106945	16.687823	123.0363	79.6435
343	41.106909	16.688015	-79.3787	-8.7999	41.106831	16.687671	133.0363	80.6435
344	41.10691	16.688049	-79.2724	-9.0194	41.106945	16.687823	131.0363	81.6435
345	41.106907	16.688087	-81.1812	-9.1343	41.106831	16.687671	141.0363	80.6435
346	41.106905	16.688125	-81.1594	-9.0805	41.106945	16.687823	137.0363	81.6435
347	41.106907	16.688173	-80.9885	-8.7323	41.106831	16.687974	147.0363	82.6435
348	41.106909	16.688222	-80.8672	-9.4353	41.106945	16.687823	144.0363	82.8935
349	41.106909	16.688265	-80.8451	-8.7382	41.106831	16.687974	154.0363	83.8935
350	41.106908	16.688309	-80.9308	-9.0304	41.106945	16.688125	144.0363	84.8935
351	41.106896	16.688342	-69.3403	-9.0126	41.106831	16.688277	154.0363	85.8935
352	41.106884	16.688374	-81.3992	-9.0235	41.106945	16.688125	144.0363	86.8935
353	41.106868	16.688406	-81.4374	-8.5219	41.106831	16.688277	154.0363	87.8935
354	41.106853	16.688438	-81.7268	-8.085	41.106716	16.688428	164.0363	88.8935
355	41.106826	16.68847	-69.7967	-8.4654	41.106831	16.688277	154.0363	89.8935
356	41.106814	16.688515	-81.6112	-7.3999	41.106716	16.688428	164.0363	90
357	41.106803	16.688547	-81.9344	-8.6966	41.106831	16.688277	154.0363	90
358	41.106789	16.688579	-81.8676	-7.2488	41.106716	16.688428	164.0363	90
359	41.106773	16.688612	-82.2468	-8.6834	41.106831	16.68858	154.0363	90
360	41.106759	16.688642	-57.1381	-6.3852	41.106831	16.68858	154.0363	90
361	41.106741	16.688663	-69.6791	-7.4137	41.106716	16.688428	149.0363	90
362	41.106724	16.688695	-68.7625	-8.2005	41.106831	16.68858	139.0363	90
363	41.106707	16.688727	-82.7549	-7.7804	41.106716	16.688428	145.0363	89
364	41.106685	16.688767	-82.5756	-8.2726	41.106831	16.68858	135.0363	90
365	41.106663	16.688807	-82.6845	-8.5082	41.106716	16.688428	142.0363	89
366	41.106646	16.688835	-83.0965	-8.6463	41.106831	16.68858	132.0363	88
367	41.106629	16.688863	-83.2866	-9.0717	41.106716	16.688428	141.0363	87
368	41.106613	16.688891	-84.1347	-8.8782	41.106602	16.68858	148.0363	88
369	41.106596	16.68892	-83.367	-8.7703	41.106487	16.688731	158.0363	89
370	41.106578	16.688945	-83.5234	-7.8467	41.106602	16.688882	148.0363	90
371	41.10656	16.688971	-83.7592	-8.7212	41.106487	16.688731	158.0363	90
372	41.106542	16.688997	-84.1385	-7.8376	41.106373	16.688882	168.0363	90
373	41.106524	16.689023	-84.4388	-8.161	41.106487	16.689034	178.0363	90
374	41.106506	16.689049	-84.23	-7.4024	41.106373	16.688882	-171.9637	90
375	41.106469	16.689094	-84.1292	-8.2217	41.106487	16.689034	178.0363	90
376	41.106498	16.68915	-84.538	-7.7473	41.106373	16.689185	-171.9637	90
377	41.106527	16.689196	-84.0005	-7.4071	41.106487	16.689034	178.0363	90
378	41.106549	16.689214	-84.0611	-7.7045	41.106373	16.689185	-171.9637	90
379	41.106602	16.68924	-67.681	-7.48	41.106487	16.689034	-179.9637	90
380	41.10665	16.689265	-68.5625	-8.542	41.106602	16.688882	170.0363	89
381	41.106676	16.689274	-70.2477	-7.8556	41.106716	16.689034	160.0363	90
382	41.106727	16.689284	-68.5985	-8.9893	41.106602	16.688882	170.0363	89
383	41.106768	16.689289	-70.5132	-7.546	41.106716	16.689034	160.0363	89.75
384	41.106812	16.689299	-70.8351	-9.6097	41.106831	16.689185	150.0363	90
385	41.106854	16.689305	-76.2433	-8.167	41.106716	16.689034	160.0363	89
386	41.106901	16.689305	-75.8563	-9.3223	41.106831	16.689185	170.0363	90
387	41.106943	16.689303	-69.2701	-8.869	41.106716	16.689034	-179.9637	89
388	41.106976	16.689295	-79.6139	-9.2317	41.106831	16.689185	-169.9637	90
389	41.107009	16.689287	-79.3229	-8.9922	41.106945	16.689337	-159.9637	90
390	41.107042	16.68928	-78.9469	-8.3633	41.10706	16.689185	-169.9637	90
391	41.107053	16.689327	-79.3708	-8.5582	41.106945	16.689337	-159.9637	90
392	41.107065	16.689374	-79.332	-8.1217	41.10706	16.689185	-169.9637	90
393	41.107073	16.689412	-79.4163	-8.3486	41.106945	16.689337	-159.9637	90
394	41.10708	16.68945	-79.4688	-8.3245	41.10706	16.689185	-169.9637	90
395	41.107091	16.689496	-79.8436	-8.4295	41.106945	16.689337	-168.9637	90
396	41.107101	16.689542	-79.9722	-8.5431	41.10706	16.689185	-178.9637	89
397	41.107109	16.689577	-79.8152	-8.7464	41.106945	16.689337	-171.9637	88
398	41.107116	16.689611	-79.4307	-8.481	41.10706	16.689488	178.0363	89
399	41.107124	16.689646	-79.6514	-8.8216	41.106945	16.689639	-171.9637	90
400	41.107131	16.68968	-79.6914	-8.4055	41.10706	16.689488	178.0363	90
401	41.107142	16.689728	-79.6443	-8.7415	41.106945	16.689639	-171.9637	90
402	41.107147	16.689794	-79.8982	-8.6648	41.10706	16.689791	-161.9637	90
403	41.107151	16.68983	-80.0952	-8.8372	41.106945	16.689639	-157.9637	89
404	41.107154	16.689866	-80.0899	-8.4636	41.10706	16.689791	-167.9637	90
405	41.107158	16.689901	-80.1698	-9.0438	41.106945	16.689942	-157.9637	90
406	41.10716	16.689933	-80.7519	-8.4349	41.10706	16.689791	-167.9637	90
407	41.107163	16.689965	-79.4523	-8.8667	41.106945	16.689942	-157.9637	90
408	41.107166	16.689997	-79.1713	-8.5817	41.10706	16.689791	-167.9637	90

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
409	41.107168	16.690028	-79.6289	-8.8358	41.106945	16.689942	-157.9637	90
410	41.107172	16.690063	-70.733	-8.5978	41.10706	16.690094	-147.9637	90
411	41.107176	16.690098	-79.4265	-8.7644	41.107174	16.690245	-137.9637	90
412	41.10718	16.690133	-78.9843	-8.4664	41.10706	16.690094	-147.9637	90
413	41.107186	16.690198	-70.293	-8.2005	41.107174	16.690245	-137.9637	90
414	41.107187	16.690239	-79.2669	-8.6335	41.10706	16.690094	-147.9637	90
415	41.107189	16.69028	-79.6914	-8.1199	41.107174	16.690245	-157.9637	90
416	41.107188	16.690323	-79.4686	-8.6799	41.10706	16.690094	-167.9637	90
417	41.107188	16.690366	-79.7839	-7.6678	41.107174	16.690245	-177.9637	90
418	41.107196	16.69041	-79.7213	-8.8975	41.10706	16.690396	-167.9637	90
419	41.107206	16.690475	-79.2714	-7.7189	41.107174	16.690245	-177.9637	90
420	41.107207	16.690514	-79.44	-8.7212	41.10706	16.690396	-167.9637	90
421	41.107209	16.690554	-79.6033	-8.1778	41.107174	16.690245	-177.9637	89
422	41.107211	16.690594	-79.2078	-8.8397	41.10706	16.690396	-172.9637	90
423	41.107214	16.690638	-77.8665	-8.7219	41.107174	16.690245	177.0363	89
424	41.107218	16.690682	-78.1861	-9.1763	41.10706	16.690396	-175.9637	88
425	41.107223	16.690722	-70.7906	-8.776	41.107174	16.690548	174.0363	89
426	41.107228	16.690762	-70.382	-8.9321	41.10706	16.690699	-175.9637	90
427	41.107213	16.690821	-70.6587	-8.0996	41.107174	16.690851	-165.9637	90
428	41.10725	16.690836	-69.3943	-8.8669	41.10706	16.690699	-155.9637	90
429	41.107287	16.690851	-66.0794	-7.2845	41.107174	16.690851	-145.9637	90
430	41.107323	16.690867	-67.824	-8.1859	41.107289	16.691002	-135.9637	90
431	41.107353	16.690879	-67.692	-7.3816	41.107174	16.690851	-134.9637	90
432	41.107383	16.690891	-67.809	-7.455	41.107289	16.691002	-124.9637	90
433	41.107413	16.690903	-67.2528	-7.869	41.107174	16.690851	-129.9637	89
434	41.107463	16.690924	-65.3646	-7.4302	41.107289	16.691002	-119.9637	90
435	41.107522	16.690939	-65.1378	-7.5568	41.107403	16.690851	-129.9637	90
436	41.107559	16.690952	-66.975	-7.676	41.107518	16.691002	-119.9637	90
437	41.107596	16.690964	-67.9937	-7.7489	41.107403	16.690851	-129.9637	89
438	41.107638	16.690976	-66.5649	-7.2355	41.107518	16.691002	-124.9637	90
439	41.107679	16.690987	-66.9583	-7.7436	41.107632	16.691153	-114.9637	90
440	41.107706	16.691005	-67.3661	-7.4314	41.107518	16.691002	-124.9637	90
441	41.107733	16.691023	-67.4585	-7.3681	41.107632	16.691153	-114.9637	90
442	41.10776	16.691041	-67.0138	-7.8975	41.107518	16.691002	-124.9637	89
443	41.107779	16.69107	-58.2484	-7.1223	41.107518	16.691002	-124.9637	89
444	41.107798	16.6911	-58.3521	-7.5365	41.107518	16.691002	-124.9637	89
445	41.107816	16.691129	-67.0363	-7.4014	41.107632	16.691153	-121.9637	90
446	41.107841	16.691165	-67.1734	-7.3763	41.107747	16.691305	-111.9637	90
447	41.107865	16.691201	-63.5698	-7.2499	41.107747	16.691305	-111.9637	90
448	41.107884	16.691227	-63.452	-7.2333	41.107747	16.691305	-111.9637	90
449	41.107902	16.691253	-63.1965	-7.2772	41.107747	16.691305	-111.9637	90
450	41.10792	16.691279	-63.4598	-7.3793	41.10809	16.691282	-111.9637	90
451	41.107938	16.691305	-63.4656	-7.5366	41.108433	16.691259	-111.9637	90
452	41.107957	16.691339	-63.6228	-7.7819	41.108776	16.691236	-111.9637	90
453	41.107976	16.691372	-63.0736	-8.0944	41.109119	16.691213	-111.9637	90
454	41.107994	16.691405	-63.8634	-8.466	41.109462	16.69119	-111.9637	90
455	41.108025	16.691416	-63.7997	-8.8166	41.109805	16.691167	-111.9637	90
456	41.108057	16.691428	-63.4575	-9.2125	41.110148	16.691144	-111.9637	90
457	41.108088	16.691439	-63.5678	-9.6442	41.110491	16.69112	-111.9637	90
458	41.108119	16.69145	-62.6512	-10.1054	41.110835	16.691097	-111.9637	90
459	41.10815	16.691462	-62.3705	-10.5858	41.111178	16.691074	-111.9637	90
460	41.108173	16.691483	-64.0129	-11.0236	41.111521	16.691051	-111.9637	90
461	41.108195	16.691504	-63.737	-11.4666	41.111864	16.691028	-111.9637	90
462	41.108218	16.691525	-63.3762	-11.9123	41.112207	16.691005	-111.9637	90
463	41.108241	16.691545	-62.8827	-12.3586	41.11255	16.690982	-111.9637	90
464	41.108263	16.691566	-67.9075	-30.005	41.11255	16.690982	82.1375	28.2814
465	41.108286	16.691587	-62.3294	-29.6706	41.11255	16.690982	82.1375	28.2814
466	41.108308	16.691608	-62.3291	-29.5296	41.11255	16.690982	82.1375	28.2814
467	41.108331	16.691629	-62.3259	-29.4556	41.11255	16.690982	82.1375	28.2814
468	41.108354	16.69165	-62.2519	-29.7215	41.112893	16.690959	82.1375	28.2814
469	41.108376	16.691671	-62.2562	-29.573	41.113236	16.690936	82.1375	28.2814
470	41.108397	16.691691	-62.3381	-29.6815	41.113579	16.690913	82.1375	28.2814
471	41.108418	16.691711	-62.2919	-29.6297	41.113922	16.69089	82.1375	28.2814
472	41.108438	16.691731	-62.6912	-30.1904	41.114266	16.690867	82.1375	28.2814
473	41.108459	16.69175	-62.6773	-30.0673	NaN	NaN	NaN	NaN
474	41.108479	16.69177	-64.1875	NaN	NaN	NaN	NaN	NaN
475	41.1085	16.69179	-63.9781	NaN	NaN	NaN	NaN	NaN
476	41.108521	16.69181	-64.0548	NaN	NaN	NaN	NaN	NaN
477	41.108541	16.69183	-63.6908	NaN	NaN	NaN	NaN	NaN
478	41.108562	16.69185	-65.5289	-36.9887	41.114503	16.690851	83.0321	24.9735
479	41.108582	16.691869	-65.6341	-38.126	41.114618	16.691002	82.0321	23.9735
480	41.108603	16.691889	-65.759	-37.8101	41.114732	16.691154	81.0321	24.7235

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
481	41.108624	16.691909	-65.4133	-37.4073	41.114847	16.691305	80.0321	24.7235
482	41.108644	16.691929	-65.68	-38.4784	41.114732	16.691154	81.0321	24.7235
483	41.108665	16.691949	-65.3227	-37.9365	41.114618	16.691305	80.0321	23.9735
484	41.108685	16.691969	-65.2476	-37.8862	41.114503	16.691457	80.0321	24.9735
485	41.108706	16.691989	-65.4781	-37.3677	41.114388	16.691608	79.0321	25.2235
486	41.108727	16.692008	-65.4203	-39.1511	41.114274	16.69176	78.0321	24.4735
487	41.108747	16.692028	-65.5111	-39.2571	41.114159	16.691911	77.0321	24.7235
488	41.108768	16.692048	-65.7002	-39.0869	41.114045	16.69176	78.0321	24.7235
489	41.108789	16.692068	-65.6436	-38.0147	41.11393	16.691608	80.0321	24.9735
490	41.108809	16.692088	-65.2441	-38.3209	41.113816	16.69176	79.0321	25.2235
491	41.10883	16.692108	-65.2484	-37.9102	41.113701	16.691911	78.0321	25.4735
492	41.10885	16.692128	-63.8631	-37.7921	41.113701	16.691911	78.0321	25.4735
493	41.108871	16.692147	-64.7198	-38.0034	41.113701	16.691911	78.0321	25.4735
494	41.108892	16.692167	-64.7631	-37.2709	41.113701	16.691911	78.0321	25.4735
495	41.108913	16.692187	-64.7011	-37.5127	41.113944	16.69159	78.0321	25.4735
496	41.108934	16.692206	-65.0758	-37.5787	41.113944	16.69159	82.7062	27.0233
497	41.108956	16.692225	-64.8875	-36.5521	41.113944	16.69159	82.7062	27.0233
498	41.108977	16.692244	-64.864	-36.6755	41.113944	16.69159	82.7062	27.0233
499	41.108999	16.692264	-64.8693	-36.552	41.113944	16.69159	82.7062	27.0233
500	41.10902	16.692283	-64.509	-36.2701	41.114187	16.691269	82.7062	27.0233
501	41.109042	16.692302	-64.9531	-37.2754	41.11443	16.690947	82.7062	27.0233
502	41.109063	16.692322	-65.0614	-38.5529	41.11443	16.690947	87.0797	25.4963
503	41.109084	16.692341	-65.0068	-38.9525	41.114316	16.690796	88.0797	24.4963
504	41.109106	16.69236	-64.66	-38.939	41.114316	16.690796	88.0797	24.4963
505	41.109127	16.69238	-64.6792	-38.8949	41.114316	16.690796	88.0797	24.4963
506	41.109149	16.692399	-65.1414	-39.184	41.114201	16.690644	90.0797	24.7463
507	41.10917	16.692418	-64.8534	-38.4945	41.114201	16.690644	90.0797	24.7463
508	41.109191	16.692438	-64.7873	-45.9197	41.114201	16.690644	90.0797	24.7463
509	41.109213	16.692457	-64.4686	-45.5638	41.114201	16.690644	90.0797	24.7463
510	41.109234	16.692476	-64.6054	-45.5032	NaN	NaN	NaN	NaN
511	41.109256	16.692495	-75.6719	-46.2136	41.114503	16.690851	88.7142	25.421
512	41.109277	16.692515	-74.8554	-46.5264	41.114388	16.6907	90.7142	24.671
513	41.109299	16.692534	-74.2193	-44.9973	41.114503	16.690548	91.7142	25.421
514	41.109318	16.692554	-74.8076	-46.6925	41.114388	16.6907	91.7142	24.921
515	41.109338	16.692573	-74.9163	-45.8821	41.114274	16.690548	92.7142	25.671
516	41.109358	16.692593	-74.9016	-45.8095	41.114159	16.6907	92.7142	25.171
517	41.109378	16.692612	-64.0847	-38.7501	41.114159	16.6907	92.7142	25.171
518	41.109398	16.692632	-63.9337	-45.9515	41.114159	16.6907	92.7142	25.171
519	41.109418	16.692652	-64.3731	-45.9005	41.114159	16.6907	92.7142	25.171
520	41.109438	16.692671	-64.3199	-45.8913	41.114485	16.690843	92.7142	25.171
521	41.109458	16.692691	-63.6357	-45.8554	NaN	NaN	NaN	NaN
522	41.109478	16.69271	-63.8442	NaN	NaN	NaN	NaN	NaN
523	41.109498	16.69273	-73.7958	-44.5171	41.114503	16.690851	92.2046	25.9685
524	41.109518	16.69275	-73.8457	-38.1657	41.114388	16.691002	91.2046	24.9685
525	41.109538	16.692769	-73.9462	-38.0343	41.114274	16.691154	90.2046	24.9685
526	41.109558	16.692789	-63.4169	-38.4059	41.114274	16.691154	90.2046	24.9685
527	41.109578	16.692808	-63.0448	-36.3484	41.114274	16.691154	90.2046	24.9685
528	41.109598	16.692828	-72.7706	-36.5333	41.114159	16.691305	90.2046	25.4685
529	41.109618	16.692848	-62.231	-38.1357	41.114159	16.691305	90.2046	25.4685
530	41.109638	16.692867	-62.4052	-37.8554	41.114159	16.691305	90.2046	25.4685
531	41.109658	16.692887	-71.9627	-37.4263	41.114274	16.691457	89.2046	26.2185
532	41.109678	16.692906	-72.4037	-37.6299	41.114388	16.691608	87.2046	25.9685
533	41.109698	16.692926	-72.3654	-37.3106	41.114503	16.691457	89.2046	25.9685
534	41.109718	16.692946	-73.2905	-37.787	41.114388	16.691608	88.2046	26.2185
535	41.109738	16.692965	-72.2761	-34.5814	41.114274	16.691457	90.2046	26.4685
536	41.109758	16.692985	-72.147	-35.6344	41.114159	16.691608	89.2046	25.7185
537	41.109778	16.693004	-73.7614	-35.1405	41.114045	16.691457	91.2046	25.7185
538	41.109798	16.693024	-62.2538	-37.4616	41.114045	16.691457	91.2046	25.7185
539	41.109818	16.693044	-62.2141	-36.5443	41.114045	16.691457	91.2046	25.7185
540	41.109838	16.693063	-73.9587	-37.2889	41.11393	16.691608	91.2046	26.2185
541	41.109858	16.693083	-72.3133	-36.1814	41.113816	16.691457	93.2046	26.2185
542	41.109878	16.693102	-72.9714	-36.1713	41.113701	16.691608	92.2046	26.4685
543	41.109898	16.693122	-72.9051	-35.0418	41.113587	16.691759	91.2046	26.7185
544	41.109918	16.693141	-71.9153	-34.2747	41.113472	16.691608	94.2046	27.7185
545	41.109938	16.693161	-73.1313	-33.7108	41.113358	16.691759	93.2046	28.7185
546	41.109958	16.693181	-74.4876	-33.0419	41.113243	16.691911	92.2046	28.9685
547	41.109978	16.6932	-72.0669	-33.4262	41.113129	16.692062	91.2046	27.9685
548	41.109998	16.69322	-74.439	-31.8135	41.113243	16.692214	89.2046	28.9685
549	41.110018	16.693239	-71.5659	-30.9936	41.113129	16.692365	88.2046	28.4685
550	41.110038	16.693259	-71.5783	-29.6934	41.113014	16.692517	86.2046	29.4685
551	41.110058	16.693279	-72.2917	-31.6564	41.1129	16.692365	89.2046	30.4685
552	41.110078	16.693298	-61.9524	-31.7152	41.1129	16.692365	89.2046	30.4685

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
553	41.110098	16.693318	-64.0865	-31.7149	41.1129	16.692365	89.2046	30.4685
554	41.110117	16.693337	-79.8597	-30.6966	41.112785	16.692517	89.2046	31.4685
555	41.110137	16.693357	-79.6506	-29.9778	41.112671	16.692668	87.2046	31.9685
556	41.110157	16.693377	-79.6434	-30.1019	41.112556	16.692819	85.2046	30.9685
557	41.110177	16.693396	-79.7367	-24.6974	41.112671	16.692971	83.2046	30.4685
558	41.110197	16.693416	-79.8277	-23.5839	41.112556	16.692819	86.2046	30.9685
559	41.110217	16.693435	-79.7705	-24.5007	41.112442	16.692971	84.2046	31.9685
560	41.110237	16.693455	-75.4106	-22.6858	41.112327	16.693122	82.2046	32.9685
561	41.110257	16.693475	-61.9138	-27.2776	41.112327	16.693122	82.2046	32.9685
562	41.110277	16.693494	-61.1387	-27.2436	41.112327	16.693122	82.2046	32.9685
563	41.110297	16.693514	-63.4616	-26.3694	41.112327	16.693122	82.2046	32.9685
564	41.110317	16.693533	-78.7665	-26.0131	41.112213	16.692971	88.2046	33.9685
565	41.110337	16.693553	-78.2977	-24.5216	41.112098	16.693122	85.2046	34.9685
566	41.110357	16.693573	-78.2556	-24.2396	41.111984	16.693274	83.2046	35.7185
567	41.110377	16.693592	-78.357	-22.8561	41.111869	16.693122	88.2046	36.7185
568	41.110397	16.693612	-74.2261	-19.4833	41.111755	16.692971	94.2046	37.7185
569	41.110417	16.693631	-75.2215	-20.3801	41.11164	16.693122	92.2046	38.7185
570	41.110437	16.693651	-74.2279	-16.8068	41.111526	16.693274	89.2046	39.7185
571	41.110457	16.69367	-74.3423	-14.8453	41.111411	16.693425	84.2046	40.7185
572	41.110477	16.69369	-75.1735	-13.7821	41.111297	16.693274	94.2046	41.7185
573	41.110497	16.69371	-78.4817	-13.1931	41.111182	16.693425	88.2046	42.7185
574	41.110517	16.693729	-74.2687	-11.3617	41.111068	16.693576	82.2046	43.7185
575	41.110537	16.693749	-78.5063	-11.7744	41.110953	16.693728	72.2046	44.7185
576	41.110557	16.693768	-74.8	-10.3856	41.110839	16.693576	82.2046	45.7185
577	41.110577	16.693788	-75.0828	-11.4364	41.110724	16.693728	80.2046	46.7185
578	41.110597	16.693808	-58.9107	-11.7597	41.110724	16.693728	80.2046	46.7185
579	41.110617	16.693827	-62.7755	-12.1786	41.110724	16.693728	80.2046	46.7185
580	41.110637	16.693847	-59.0268	-12.7022	41.110724	16.693728	80.2046	46.7185
581	41.110657	16.693866	-59.2964	-13.3389	41.111022	16.693501	80.2046	46.7185
582	41.110677	16.693886	-59.5102	-14.0955	41.111319	16.693275	80.2046	46.7185
583	41.110698	16.693906	-59.443	-15.0464	41.111617	16.693048	80.2046	46.7185
584	41.11072	16.693927	-58.8261	-16.1478	41.111915	16.692821	80.2046	46.7185
585	41.110741	16.693947	-60.8465	-17.409	41.112213	16.692595	80.2046	46.7185
586	41.110763	16.693968	-61.0211	-18.8431	41.11251	16.692368	80.2046	46.7185
587	41.110784	16.693988	-60.7332	-20.4689	41.112808	16.692141	80.2046	46.7185
588	41.110806	16.694009	-63.5988	-22.3151	41.113106	16.691915	80.2046	46.7185
589	41.110827	16.694029	-74.446	-35.4594	41.113106	16.691915	113.0416	33.7853
590	41.110849	16.69405	-74.4528	-36.5554	41.11322	16.692066	108.0416	32.7853
591	41.11087	16.69407	-73.882	-36.1054	41.113106	16.692218	108.0416	31.7853
592	41.110891	16.694091	-74.0316	-34.6692	41.112991	16.692369	107.0416	32.5353
593	41.110913	16.694111	-73.9377	-32.9708	41.112877	16.69252	107.0416	33.5353
594	41.110934	16.694131	-78.03	-33.0605	41.112762	16.692672	107.0416	32.5353
595	41.110956	16.694152	-60.4371	-22.281	41.112762	16.692672	107.0416	32.5353
596	41.110977	16.694172	-61.4717	-22.105	41.112762	16.692672	107.0416	32.5353
597	41.110999	16.694193	-61.4802	-22.3407	41.112762	16.692672	107.0416	32.5353
598	41.111102	16.694213	-61.5463	-24.9794	41.113032	16.69239	107.0416	32.5353
599	41.111045	16.694234	-62.109	-22.4953	41.113301	16.692108	107.0416	32.5353
600	41.111107	16.694254	-58.268	-22.3276	41.113571	16.691826	107.0416	32.5353
601	41.111095	16.694275	-57.8752	-22.1721	41.11384	16.691544	107.0416	32.5353
602	41.111112	16.694295	-58.2592	-21.6206	41.114109	16.691263	107.0416	32.5353
603	41.111141	16.694266	-57.9432	-21.758	41.114379	16.690981	107.0416	32.5353
604	41.111161	16.694237	-58.6435	-21.7934	NaN	NaN	NaN	NaN
605	41.111182	16.694207	-58.5555	NaN	NaN	NaN	NaN	NaN
606	41.111202	16.694178	-58.6063	NaN	NaN	NaN	NaN	NaN
607	41.111226	16.694148	-57.7732	NaN	NaN	NaN	NaN	NaN
608	41.11125	16.694118	-58.1207	NaN	NaN	NaN	NaN	NaN
609	41.111274	16.694087	-58.0772	NaN	NaN	NaN	NaN	NaN
610	41.111291	16.694059	-57.9069	NaN	NaN	NaN	NaN	NaN
611	41.111308	16.69403	-57.8077	NaN	NaN	NaN	NaN	NaN
612	41.111325	16.694002	-58.8025	NaN	NaN	NaN	NaN	NaN
613	41.111342	16.693974	-59.8601	NaN	NaN	NaN	NaN	NaN
614	41.111359	16.693946	-60.4246	NaN	NaN	NaN	NaN	NaN
615	41.111376	16.693917	-60.1253	NaN	NaN	NaN	NaN	NaN
616	41.111393	16.693889	-58.3885	NaN	NaN	NaN	NaN	NaN
617	41.111408	16.693864	-58.2982	NaN	NaN	NaN	NaN	NaN
618	41.111422	16.693839	-59.8498	NaN	NaN	NaN	NaN	NaN
619	41.111437	16.693814	-59.7636	NaN	NaN	NaN	NaN	NaN
620	41.111451	16.693789	-58.5895	NaN	NaN	NaN	NaN	NaN
621	41.111466	16.693764	-59.8378	NaN	NaN	NaN	NaN	NaN
622	41.11148	16.693739	-59.5012	NaN	NaN	NaN	NaN	NaN
623	41.111494	16.693715	-59.4667	NaN	NaN	NaN	NaN	NaN
624	41.111509	16.69369	-58.3543	NaN	NaN	NaN	NaN	NaN

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
625	41.111523	16.693665	-58.0196	NaN	NaN	NaN	NaN	NaN
626	41.111538	16.69364	-56.9184	NaN	NaN	NaN	NaN	NaN
627	41.111552	16.693615	-56.9577	NaN	NaN	NaN	NaN	NaN
628	41.111567	16.69359	-56.6246	NaN	NaN	NaN	NaN	NaN
629	41.111581	16.693565	-57.7904	NaN	NaN	NaN	NaN	NaN
630	41.111595	16.69354	-57.6646	NaN	NaN	NaN	NaN	NaN
631	41.11161	16.693515	-56.8234	NaN	NaN	NaN	NaN	NaN
632	41.111624	16.69349	-58.0878	NaN	NaN	NaN	NaN	NaN
633	41.111639	16.693466	-60.8207	NaN	NaN	NaN	NaN	NaN
634	41.111653	16.693441	-58.9303	NaN	NaN	NaN	NaN	NaN
635	41.111668	16.693416	-59.0495	NaN	NaN	NaN	NaN	NaN
636	41.111682	16.693391	-59.4727	NaN	NaN	NaN	NaN	NaN
637	41.111697	16.693366	-58.9621	NaN	NaN	NaN	NaN	NaN
638	41.111711	16.693341	-58.8005	NaN	NaN	NaN	NaN	NaN
639	41.111725	16.693316	-58.1358	NaN	NaN	NaN	NaN	NaN
640	41.11174	16.693291	-59.1861	NaN	NaN	NaN	NaN	NaN
641	41.111754	16.693266	-58.2096	NaN	NaN	NaN	NaN	NaN
642	41.111769	16.693241	-58.1885	NaN	NaN	NaN	NaN	NaN
643	41.111783	16.693216	-60.8381	NaN	NaN	NaN	NaN	NaN
644	41.111798	16.693192	-57.5894	NaN	NaN	NaN	NaN	NaN
645	41.111812	16.693167	-57.9241	NaN	NaN	NaN	NaN	NaN
646	41.111826	16.693142	-58.4913	NaN	NaN	NaN	NaN	NaN
647	41.111841	16.693117	-58.1639	NaN	NaN	NaN	NaN	NaN
648	41.111855	16.693092	-57.8362	NaN	NaN	NaN	NaN	NaN
649	41.11187	16.693067	-58.1365	NaN	NaN	NaN	NaN	NaN
650	41.111884	16.693042	-59.3037	NaN	NaN	NaN	NaN	NaN
651	41.111899	16.693017	-58.9325	NaN	NaN	NaN	NaN	NaN
652	41.111913	16.692991	-60.2113	NaN	NaN	NaN	NaN	NaN
653	41.111928	16.692965	-56.5443	NaN	NaN	NaN	NaN	NaN
654	41.111943	16.692939	-59.8984	NaN	NaN	NaN	NaN	NaN
655	41.111957	16.692913	-59.0598	NaN	NaN	NaN	NaN	NaN
656	41.111972	16.692887	-56.3105	NaN	NaN	NaN	NaN	NaN
657	41.111987	16.692861	-57.9813	NaN	NaN	NaN	NaN	NaN
658	41.112001	16.692835	-59.6564	NaN	NaN	NaN	NaN	NaN
659	41.112016	16.692809	-55.7627	NaN	NaN	NaN	NaN	NaN
660	41.112031	16.692783	-58.0774	NaN	NaN	NaN	NaN	NaN
661	41.112045	16.692757	-55.9002	NaN	NaN	NaN	NaN	NaN
662	41.11206	16.692731	-55.877	NaN	NaN	NaN	NaN	NaN
663	41.112075	16.692705	-55.3412	NaN	NaN	NaN	NaN	NaN
664	41.11209	16.692679	-53.4201	NaN	NaN	NaN	NaN	NaN
665	41.112104	16.692653	-53.3962	NaN	NaN	NaN	NaN	NaN
666	41.112118	16.692628	-53.3855	NaN	NaN	NaN	NaN	NaN
667	41.112131	16.692602	-53.482	NaN	NaN	NaN	NaN	NaN
668	41.112145	16.692577	-58.5453	NaN	NaN	NaN	NaN	NaN
669	41.112158	16.692552	-55.0385	NaN	NaN	NaN	NaN	NaN
670	41.112172	16.692527	-55.012	NaN	NaN	NaN	NaN	NaN
671	41.112185	16.692501	-57.4673	NaN	NaN	NaN	NaN	NaN
672	41.112199	16.692476	-55.4562	NaN	NaN	NaN	NaN	NaN
673	41.112213	16.692451	-57.2406	NaN	NaN	NaN	NaN	NaN
674	41.112226	16.692426	-57.7028	NaN	NaN	NaN	NaN	NaN
675	41.11224	16.6924	-55.7203	NaN	NaN	NaN	NaN	NaN
676	41.112253	16.692375	-56.3567	NaN	NaN	NaN	NaN	NaN
677	41.112267	16.69235	-55.2183	NaN	NaN	NaN	NaN	NaN
678	41.11228	16.692325	-53.8246	NaN	NaN	NaN	NaN	NaN
679	41.112294	16.6923	-57.7413	NaN	NaN	NaN	NaN	NaN
680	41.112307	16.692274	-53.6452	NaN	NaN	NaN	NaN	NaN
681	41.112321	16.692249	-55.681	NaN	NaN	NaN	NaN	NaN
682	41.112335	16.692224	-51.9139	NaN	NaN	NaN	NaN	NaN
683	41.112349	16.692198	-55.1723	NaN	NaN	NaN	NaN	NaN
684	41.112364	16.692172	-57.1602	NaN	NaN	NaN	NaN	NaN
685	41.112379	16.692147	-56.5997	NaN	NaN	NaN	NaN	NaN
686	41.112394	16.692121	-54.3613	NaN	NaN	NaN	NaN	NaN
687	41.112409	16.692095	-54.6843	NaN	NaN	NaN	NaN	NaN
688	41.112423	16.692069	-53.6037	NaN	NaN	NaN	NaN	NaN
689	41.112438	16.692044	-56.5009	NaN	NaN	NaN	NaN	NaN
690	41.112453	16.692018	-55.0483	NaN	NaN	NaN	NaN	NaN
691	41.112468	16.691992	-56.4649	NaN	NaN	NaN	NaN	NaN
692	41.112483	16.691966	-50.33	NaN	NaN	NaN	NaN	NaN
693	41.112498	16.691941	-49.5587	NaN	NaN	NaN	NaN	NaN
694	41.112512	16.691915	-50.9246	NaN	NaN	NaN	NaN	NaN
695	41.112527	16.691889	-50.9429	NaN	NaN	NaN	NaN	NaN
696	41.112542	16.691863	-48.6549	NaN	NaN	NaN	NaN	NaN

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RX	Lat RX	Lon RX	P_{senza} [dBm]	P_{con} [dBm]	Lat drone	Lon drone	Azimuth [deg]	Elevation [deg]
697	41.112557	16.691837	-51.4651	NaN	NaN	NaN	NaN	NaN
698	41.112572	16.691812	-60.1234	NaN	NaN	NaN	NaN	NaN
699	41.112587	16.691787	-60.1982	NaN	NaN	NaN	NaN	NaN
700	41.112603	16.691762	-60.4791	NaN	NaN	NaN	NaN	NaN
701	41.112618	16.691736	-61.0532	NaN	NaN	NaN	NaN	NaN
702	41.112633	16.691711	-61.1291	NaN	NaN	NaN	NaN	NaN
703	41.112649	16.691686	-60.6108	NaN	NaN	NaN	NaN	NaN
704	41.112664	16.69166	-62.606	NaN	NaN	NaN	NaN	NaN
705	41.112679	16.691635	-62.4593	NaN	NaN	NaN	NaN	NaN
706	41.112695	16.69161	-62.7269	NaN	NaN	NaN	NaN	NaN
707	41.11271	16.691585	-50.2028	NaN	NaN	NaN	NaN	NaN
708	41.112725	16.691559	-50.2121	NaN	NaN	NaN	NaN	NaN
709	41.11274	16.691534	-51.3562	NaN	NaN	NaN	NaN	NaN
710	41.112756	16.691509	-51.3102	NaN	NaN	NaN	NaN	NaN
711	41.112771	16.691484	-50.4319	NaN	NaN	NaN	NaN	NaN
712	41.112799	16.691484	-50.584	NaN	NaN	NaN	NaN	NaN
713	41.112828	16.691484	-50.588	NaN	NaN	NaN	NaN	NaN
714	41.112856	16.691484	-50.4455	NaN	NaN	NaN	NaN	NaN
715	41.112885	16.691484	-49.9632	NaN	NaN	NaN	NaN	NaN
716	41.112913	16.691485	-49.7651	NaN	NaN	NaN	NaN	NaN
717	41.112942	16.691485	-49.6167	NaN	NaN	NaN	NaN	NaN
718	41.11297	16.691485	-49.0493	NaN	NaN	NaN	NaN	NaN
719	41.112999	16.691485	-48.8935	NaN	NaN	NaN	NaN	NaN
720	41.113027	16.691485	-48.7386	NaN	NaN	NaN	NaN	NaN
721	41.113055	16.691486	-47.4855	NaN	NaN	NaN	NaN	NaN
722	41.113084	16.691486	-47.3087	NaN	NaN	NaN	NaN	NaN
723	41.113112	16.691486	-47.1304	NaN	NaN	NaN	NaN	NaN
724	41.113141	16.691486	-46.9529	NaN	NaN	NaN	NaN	NaN
725	41.113169	16.691486	-47.2122	NaN	NaN	NaN	NaN	NaN
726	41.113198	16.691487	-47.0691	NaN	NaN	NaN	NaN	NaN
727	41.113226	16.691487	-46.9275	NaN	NaN	NaN	NaN	NaN
728	41.113255	16.691487	-43.5306	NaN	NaN	NaN	NaN	NaN
729	41.113283	16.691487	-43.3553	NaN	NaN	NaN	NaN	NaN
730	41.113311	16.691487	-40.7318	NaN	NaN	NaN	NaN	NaN
731	41.11334	16.691488	-39.5448	NaN	NaN	NaN	NaN	NaN
732	41.113368	16.691488	-39.2661	NaN	NaN	NaN	NaN	NaN
733	41.113397	16.691488	-38.9848	NaN	NaN	NaN	NaN	NaN
734	41.113425	16.691488	-38.7007	NaN	NaN	NaN	NaN	NaN
735	41.113454	16.691488	-36.2947	NaN	NaN	NaN	NaN	NaN
736	41.113482	16.691489	-35.9388	NaN	NaN	NaN	NaN	NaN
737	41.113511	16.691489	-35.529	NaN	NaN	NaN	NaN	NaN
738	41.113538	16.691503	-34.9283	NaN	NaN	NaN	NaN	NaN
739	41.113565	16.691516	-34.3292	NaN	NaN	NaN	NaN	NaN
740	41.113592	16.69153	-33.8052	NaN	NaN	NaN	NaN	NaN
741	41.113619	16.691544	-33.504	NaN	NaN	NaN	NaN	NaN
742	41.113646	16.691558	-33.2065	NaN	NaN	NaN	NaN	NaN
743	41.113673	16.691572	-32.9165	NaN	NaN	NaN	NaN	NaN
744	41.113701	16.691585	-32.6175	NaN	NaN	NaN	NaN	NaN
745	41.113729	16.691579	-32.1726	NaN	NaN	NaN	NaN	NaN
746	41.113758	16.691573	-31.7186	NaN	NaN	NaN	NaN	NaN
747	41.113787	16.691567	-31.2507	NaN	NaN	NaN	NaN	NaN
748	41.113815	16.691561	-30.773	NaN	NaN	NaN	NaN	NaN
749	41.113844	16.691555	-30.3576	NaN	NaN	NaN	NaN	NaN
750	41.113873	16.691549	-29.7741	NaN	NaN	NaN	NaN	NaN
751	41.113902	16.691543	-29.4456	NaN	NaN	NaN	NaN	NaN
752	41.11393	16.691537	-28.9548	NaN	NaN	NaN	NaN	NaN
753	41.113959	16.69153	-28.4374	NaN	NaN	NaN	NaN	NaN
754	41.113988	16.691524	-27.9345	NaN	NaN	NaN	NaN	NaN
755	41.114016	16.691518	-27.4575	NaN	NaN	NaN	NaN	NaN
756	41.114045	16.691512	-26.9789	NaN	NaN	NaN	NaN	NaN
757	41.114074	16.691506	-26.4731	NaN	NaN	NaN	NaN	NaN
758	41.114102	16.6915	-25.9718	NaN	NaN	NaN	NaN	NaN
759	41.114131	16.691494	-25.3975	NaN	NaN	NaN	NaN	NaN
760	41.11416	16.691488	-24.8184	NaN	NaN	NaN	NaN	NaN
761	41.114189	16.691481	-39.9803	NaN	NaN	NaN	NaN	NaN